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Explainable Artificial Intelligence in Sports Analytics

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Introduction

Over the past decade, football analytics practices have changed due to increased data availability and analytical adoption.(Lolli et al., 2025) The widespread use of high-resolution event data, generated through modern tracking systems and event-logging platforms, has enabled clubs, analysts, and governing bodies to quantitatively examine team behaviour, match dynamics, and patterns.(Pappalardo et al., 2019) As a result, data-driven decision-making has been increasingly adopted for tactical preparation, player evaluation, and long-term strategic planning in professional football.(*Football's Evolution In The Digital Era: How Tech Is Transforming The Game*, n.d.)

Within this evolving landscape, machine learning (ML) has emerged as a methodological framework for extracting meaningful patterns from complex, high-dimensional football data. Supervised learning models are now employed for tasks such as match outcome prediction, expected goals (xG) modelling, player valuation, and tactical profiling.(Moya et al., 2025) Despite their strong predictive performance, these approaches, particularly ensemble models and deep learning techniques, are often criticised for their limited interpretability.(Kranzinger et al., 2025) In applied football environments, where analytical insights must be communicated effectively to coaches and technical staff, predictive accuracy alone is insufficient. Indeed, the opacity of black-box models can make it difficult for coaches and analysts to understand or trust the results. (Jia et al., 2025) Understanding which performance factors drive a model's output, and why specific outcomes are predicted, is equally important.

This challenge has contributed to growing interest in Explainable Artificial Intelligence (XAI) within sports analytics. XAI methods aim to reconcile predictive performance with transparency by producing interpretable models or explanations that reveal the contribution of individual features. (Kranzinger et al., 2025)In football analytics, explainability enables analysts to move beyond black-box predictions and instead identify clear, interpretable tactical patterns such as pressing intensity, spatial dominance, or passing behaviour associated with successful outcomes.(Stafylidis et al., 2024)Interpretable models not only foster trust in analytical systems but also enhance their practical usability in professional sporting contexts.(Kranzinger et al., 2025)

Another critical issue in football analytics is the heterogeneity of competitions and playing styles. Tactical behaviors vary substantially across leagues due to differences in game tempo, physical demands, refereeing standards, and cultural approaches to the sport. For example, research indicates that the English Premier League tends toward a direct, up-tempo style; Italy's Serie A emphasizes defensive tactical rigor; and Spain's La Liga favors a more possession-based, technical game, reflecting distinct cultural and historical philosophies in each competition(*English Premier League, Spain's La Liga and Italy's Serie's A - What's Different?*, n.d.). Consequently, models trained on aggregated multi-league data may capture league-specific artifacts rather than general football principles if these differences are not accounted for. In addition, team performance is not static over time; it evolves due to factors like injuries, player transfers, fatigue accumulation, and tactical adjustments. Empirical studies have shown, for instance, that high player availability and lower injury burdens are strongly associated with better team performance and higher league standings(Häggglund et

al., 2013). Therefore, robust analytical pipelines must handle both cross-competition variability and temporal dynamics in order to provide reliable, realistic evaluations.

In response to these challenges, this thesis builds upon a curricular internship conducted at Rulx Innovation Labs and focuses on the application of explainable machine learning techniques to football event data. The work integrates data engineering, feature construction, and model evaluation into a unified analytical workflow, with the explicit aim of understanding team structure and quantifying the importance of performance parameters. The analysis relies on open-access football data and employs the Logic Learning Machine (LLM) algorithm implemented within the Rulx platform, which generates transparent, rule-based models in the form of human-readable IF–THEN rules.(Muselli et al., n.d.)

Football event data offer a detailed representation of match activity by recording every on-ball actionsuch as passes, shots, duels, and interceptionstogether with their spatial and temporal attributes. Among publicly available datasets, the Wyscout Open Data repository has become a widely used benchmark due to its coverage of multiple top European leagues and its consistent annotation standards. However, raw event data are not directly suitable for machine learning applications. Meaningful analysis requires transforming low-level events into higher-level features that reflect tactical concepts.

To achieve this, football analytics commonly rely on aggregated metrics and spatial indicators. Measures such as Passes Per Defensive Action (PPDA) and Expected Goals (xG) provide intuitive quantifications of pressing intensity and chance quality, respectively, and form a natural bridge between data-driven modelling and football domain knowledge. Spatial context further enhances interpretability: dividing the pitch into functional zones—defensive, middle, and attacking thirds allows analysts to capture territorial control and attacking emphasis in a tactically meaningful way. Zone-based features have been shown to improve both model performance and interpretability (Power et al., 2017; Bransen & Van Haaren, 2019).

This thesis adopts such an approach by transforming raw JSON event files into a structured relational database, enabling systematic feature extraction through structured queries. Match- and team-level statistics are enriched with combined metrics and spatial indicators, resulting in a comprehensive and interpretable representation of team performance suitable for explainable machine learning.

Explainability is treated as a central design principle throughout the project. Feature selection is performed using ROC–AUC-based ranking to retain only the most informative attributes, while model evaluation is conducted through sliding-window validation to respect the temporal structure of football seasons and mimic real-world forecasting scenarios.

The primary objective of the thesis is to analyse football team structure and quantify the importance of performance parameters using an explainable machine learning framework. Specifically, the project aims to design a complete football analytics pipeline, identify and rank influential tactical features, extract transparent rule-based models using the Logic Learning Machine, evaluate performance under realistic temporal conditions across multiple European

leagues, and investigate cross-competition differences in tactical patterns and their generalisability.

Data Acquisition and Integration

Data acquisition and integration constitute a foundational stage in football analytics pipelines, particularly when machine learning methods are applied to analyse team structure and performance determinants. The quality, reliability, and organisation of data directly influence both the predictive accuracy and the explanatory power of analytical models. In football analytics, this challenge is intensified by the complexity of event-based datasets, the hierarchical relationships between competitions, matches, teams, players, and events, and the need to preserve spatial and temporal detail to accurately capture tactical behaviour.

This section outlines the data acquisition and integration strategy adopted in the thesis, following practices in sports data engineering. The overall objective is to transform raw football event data into a structured, machine-learning-ready representation that supports both predictive modelling and interpretability. The workflow encompasses data source selection, parsing and restructuring of hierarchical JSON files, and the construction of a relational database that serves as the backbone of the analytical pipeline.

Event-based football data represent one of the most widely used data modalities in modern sports analytics. Unlike tracking data, which records continuous player movements, event data captures discrete on-ball actions such as passes, shots, duels, fouls, and interceptions, together with contextual attributes including time, location, and player identity. This data format offers a favourable balance between availability, interpretability, and computational efficiency, making it particularly suitable for large-scale studies and explainable machine learning applications (Pappalardo et al., 2019). Prior research has shown that event data are especially effective for modelling team-level tactical behaviour, as they can be aggregated into meaningful indicators related to passing structure, pressing intensity, and attacking efficiency. Moreover, event-derived features align well with human football expertise, reinforcing their suitability for explainable analytical frameworks. (Pappalardo et al., 2019)

The primary data source used in this project is the Wyscout Open Data repository, a publicly available dataset that has become a benchmark in academic football analytics. The dataset provides detailed event-level information across multiple top European leagues, including the English Premier League, Italian Serie A, Spanish La Liga, German Bundesliga, and French Ligue 1, as well as major international tournaments. Data are distributed in structured JSON files covering matches, teams, players, and events. This dataset was selected due to its multi-competition coverage, high annotation granularity, and widespread adoption in peer-reviewed research, which supports methodological comparability and reproducibility. (Pappalardo et al., 2019)

Within the project, multiple competitions were integrated into a unified analytical framework to capture variability in tactical styles and match contexts. Particular attention was given to competitions with complete and homogeneous season structures, enabling the later use of temporally consistent validation strategies, such as sliding-window evaluation. This choice

supports realistic performance assessment and mitigates biases arising from temporal leakage.

A key technical challenge addressed in this chapter is the hierarchical nature of football event data stored in JSON format. While JSON is well suited for representing nested relationships such as events belonging to matches and competitions, it is not directly compatible with most statistical and machine learning tools, which require tabular or relational representations. Football event JSON files typically contain deeply nested attributes, including action types, sub-events, spatial coordinates, and qualifiers. Without systematic parsing, these structures remain inaccessible for analysis.

To overcome this limitation, a dedicated parsing workflow was implemented using Python. Following recent recommendations in sports data engineering, the approach emphasised early normalisation of raw data into analytically meaningful entities. JSON parsing libraries were used to navigate nested structures, while data manipulation tools flattened hierarchical objects into structured tables. Explicit key management ensured that relationships between matches, teams, players, and events were preserved. Event-level data were decomposed into atomic variables describing action characteristics, spatial location, and timing, while retaining references to higher-level entities.

An important design decision was the relational decomposition of the dataset. Instead of maintaining a single wide table, the data were organised into distinct entities: matches, teams, players, and events, reflecting principles of database normalisation. This structure improves consistency by avoiding redundant storage of shared information, enhances scalability by facilitating the addition of new competitions or seasons, and enables flexible aggregation through relational queries. The resulting data model closely mirrors the conceptual organisation adopted in professional football analytics systems.

Following parsing, the data were stored in a relational SQLite database. Relational databases are particularly suitable for football analytics due to their structured schemas, support for complex queries, and enforcement of data integrity. SQLite was chosen as a lightweight, file-based database engine that integrates seamlessly with Python workflows while remaining portable and reproducible. Despite its simplicity, SQLite provides sufficient performance for datasets of the scale considered in this research and supports standard SQL operations essential for feature extraction.

The database schema was designed to reflect the logical structure of football competitions and events. Core tables included matches, teams, players, and events, each defined by primary keys and linked through foreign key relationships to ensure referential integrity. This design allows every event to be traced back to its corresponding match and team, supporting consistent aggregation at multiple analytical levels.

Once populated, the SQLite database functioned as the central integration layer of the project. SQL queries were used to compute match and team-level statistics, aggregate event counts, and derive advanced indicators that fed directly into the machine learning pipeline. By centralising data access within a relational database, feature extraction became

transparent, reproducible, and compatible with downstream analytical tools, including the explainable modelling framework used in later chapters.

In summary, this section presented a robust data acquisition and integration workflow that transforms raw football event data into a structured, analysis-ready format. By combining systematic JSON parsing, relational normalisation, and database-driven feature extraction, the proposed approach ensures data quality, analytical flexibility, and methodological transparency. This integrated data foundation is essential for the explainable machine learning analyses developed in subsequent chapters.

Feature Engineering

Feature engineering is a central component of football analytics pipelines, particularly when the objective is to analyse team structure and quantify the importance of performance parameters using machine learning. Unlike domains characterised by massive, clean datasets, football analytics typically operates under conditions of limited sample size, heterogeneous match contexts, and high observational noise. These constraints significantly limit the effectiveness of data-hungry approaches such as deep neural networks, which generally require large training sets and stable input representations to generalise reliably.

Recent research in sports analytics highlights that, in small to medium sample regimes, carefully designed, domain-informed features often outperform automatically learned representations in both predictive performance and interpretability. Football event data are inherently noisy: not all recorded actions are equally informative, and many low-level events have limited tactical meaning when considered in isolation. As a result, directly modelling raw event counts or relying on deep learning-based feature extraction risks amplifying noise rather than uncovering meaningful patterns.

In response to these challenges, this thesis adopts a feature-centric modelling strategy. Rather than attempting to learn latent representations through complex architectures, the pipeline focuses on transforming raw event data into structured, aggregated, and interpretable features that encode tactical information at the team level. This design choice aligns naturally with the explainable machine learning framework implemented in the Rulex platform and supports the extraction of transparent rules linking team behaviour to match outcomes. The feature engineering pipeline is organised into three main components: SQL-based statistical features, composite tactical indicators, and zone-based spatial features.

The first stage of feature engineering involves the extraction of aggregated statistical features from the relational database described in the previous chapter. While individual events provide fine-grained information, machine learning models benefit from representations that summarise behaviour over meaningful temporal units, such as matches. Aggregation reduces variance, mitigates annotation noise, and produces features that are more stable across games, teams, and competitions.

From a methodological perspective, aggregation is particularly important given the limited number of matches available per season and competition. Modelling event sequences directly would result in an unfavourable ratio between feature dimensionality and sample size,

increasing the risk of overfitting. Aggregated statistics, therefore act as a form of implicit regularisation, reducing model complexity while preserving essential information about team behaviour.

Using the SQLite database as a central integration layer, SQL queries were employed to compute match-level and team-level statistics. These features include counts of passes, shots, crosses, duels, fouls, and defensive actions such as tackles and interceptions, as well as breakdowns of pass types and symmetric statistics for home and away teams. The use of SQL provides several advantages: it ensures full reproducibility by defining each feature through explicit queries, allows efficient aggregation over large event tables without unnecessary data duplication, and enforces a clear separation between data storage and feature computation. These SQL-based features form the backbone of the analytical dataset and provide a baseline quantitative description of team activity.

Despite their usefulness, purely statistical features exhibit important limitations. Raw event counts do not capture context, quality, or intent. For example, a high volume of passes may indicate either effective ball circulation or unproductive possession, while a large number of defensive actions may reflect tactical aggressiveness or prolonged pressure from the opponent. Such ambiguities limit the explanatory value of simple counts and motivate the inclusion of higher-level tactical indicators.

The second stage of the pipeline, therefore, introduces composite tactical indicators designed to encode established football concepts. Tactical indicators translate combinations of raw events into interpretable measures of team behaviour and embed domain knowledge directly into the feature space. This approach is particularly advantageous in noisy and data-limited settings, as it constrains the model to learn from features with clear semantic meaning. Among the many proposed metrics, Passes Per Defensive Action (PPDA) and Expected Goals (xG) were selected due to their widespread adoption and empirical validation in both academic research and professional practice (Power et al., 2017; Bransen & Van Haaren, 2019).

PPDA is a measure of pressing intensity, defined as the number of passes an opponent is allowed to complete before a team performs a defensive action. Lower PPDA values correspond to more aggressive pressing strategies, while higher values indicate a more passive defensive approach (Stafylidis et al., 2024). In this project, PPDA was computed using SQL queries that aggregate opponent passes and defensive actions at the match level. The metric captures a fundamental structural aspect of team behaviour: how actively a team attempts to disrupt the opponent's build-up play. PPDA is particularly well suited for explainable modelling, as it corresponds to a clear and widely understood tactical concept and has been shown to correlate with territorial dominance and match outcomes across leagues.

Expected Goals (xG) provide a complementary perspective by quantifying the quality of goal-scoring opportunities. Unlike raw shot counts, xG estimates the probability that a given shot results in a goal based on contextual factors such as shot distance, angle, and situation. In the present project, xG values were assigned to shot events using their spatial and contextual attributes and then aggregated at the team level. This allows the model to distinguish

between teams that generate many low-quality attempts and those that create fewer but more dangerous chances. Methodologically, xG serves as a noise-reducing feature by smoothing the randomness inherent in goal scoring, which is especially valuable in small-sample settings where match outcomes are subject to high variance.

While tactical indicators capture qualitative aspects of play, they do not fully describe where actions occur on the pitch. Spatial context is essential for understanding team structure, as identical actions can have very different tactical meanings depending on their location. However, directly modelling continuous spatial coordinates would substantially increase feature dimensionality and sensitivity to noise. To address this trade-off, the pipeline adopts a zone-based spatial abstraction.

The football pitch was discretised into three main zones: defensive third, middle third, and attacking third. Using event coordinates, each action was assigned to one of these zones, and zone-specific features were computed accordingly. Examples include the number of high passes in the attacking third or crosses originating from the defensive zone. These features capture how teams distribute their actions spatially and provide insight into tactical orientation, risk-taking behaviour, and territorial emphasis.

Zone-based features are particularly effective for explainable machine learning. By aggregating spatial information into intuitive regions, they reduce noise while yielding attributes that are immediately interpretable. Rules involving features such as “passes in the attacking third” or “defensive actions in the middle third” can be directly translated into tactical statements, unlike abstract latent features produced by deep learning models. Empirically, these features proved highly informative in the Rulex analysis, consistently appearing among the top-ranked attributes across multiple competitions.

In summary, this section presented a feature engineering pipeline specifically designed to address the constraints of football analytics: limited data availability, noisy observations, and the need for interpretability. By combining SQL-based statistical features, composite tactical indicators such as PPDA and xG, and zone-based spatial abstractions, the pipeline constructs a compact yet expressive representation of team performance. Rather than relying on deep learning approaches ill-suited to the data regime, the methodology embeds football knowledge directly into the feature space. This design enables robust and transparent, and explainable machine learning analyses, providing a solid foundation for the modelling and evaluation stages developed in the subsequent chapter.

Machine Learning models

This section presents the complete technical stack adopted in the thesis, describing the machine learning models, analytical workflows, and explainability mechanisms used to analyse football team structure and assess the relative importance of performance parameters. Unlike purely predictive studies that prioritise maximising classification accuracy, the primary objective of this work is knowledge extraction. As a result, model selection and tool design are guided by interpretability, robustness under limited data conditions, and methodological transparency.

Football analytics constitutes a challenging modelling environment characterised by three fundamental constraints. First, the effective sample size is intrinsically limited. Even a full domestic league season typically contains fewer than 400 matches, and once realistic temporal validation strategies are applied, the amount of data available for training at each iteration is substantially reduced. Second, football event data are inherently noisy: recorded actions differ widely in tactical relevance and are influenced by contextual factors such as match state, opposition quality, and refereeing decisions. Third, applied football analytics requires human interpretability, as analytical insights must be communicated to coaches, analysts, and decision-makers who reason in tactical and conceptual terms rather than abstract statistical representations. As noted in sports science literature, the lack of interpretability in “black-box” ML models poses a barrier to their acceptance and practical application among coaches and analysts (Kranzinger et al., 2025).

Deep neural networks generally require large volumes of data to learn stable representations and, even when successful, tend to produce latent features that are difficult to interpret in football-specific terms. Empirical evidence and recent literature suggest that, under small-sample and noisy conditions, such models are prone to overfitting and limited generalisation (Yeung et al., 2023). Consequently, this thesis adopts a model-centric and explainability-driven approach, combining classical machine learning methods, ensemble techniques, and inherently interpretable algorithms within a unified analytical framework.

The technical stack is structured around three main pillars. First, multiple machine learning models are employed for prediction and comparative evaluation. Second, the Rulx Factory platform orchestrates feature selection, training, validation, and rule extraction in a consistent and reproducible manner. Third, Explainable Artificial Intelligence (XAI) principles guide both model choice and interpretation, ensuring that analytical outputs can be translated into football-relevant knowledge.

The selection of machine learning models deliberately spans different modelling paradigms, from simple and inherently interpretable algorithms to more complex ensemble-based approaches. This diversity enables a systematic comparison between predictive performance, robustness under limited data, and interpretability. The models considered are Decision Trees, Random Forests (Biau & Scornet, 2016), Extreme Gradient Boosting (XGBoost), Explainable Boosting Machines (EBM), and the Logic Learning Machine (LLM). All models are trained on the same engineered feature space derived from football event data and tactical indicators, and they are evaluated using identical temporal validation schemes based on sliding windows. This design ensures methodological consistency and prevents confounding effects due to differences in preprocessing or evaluation strategy.

Decision Trees represent the simplest and most transparent model considered in the study. A decision tree is a non-parametric supervised learning algorithm that recursively partitions the feature space into a hierarchical structure of decision rules. Each internal node corresponds to a conditional test on a feature, and each leaf node produces a prediction. The resulting structure can be interpreted as a set of nested IF–THEN rules. In football analytics, this property is particularly appealing, as decision paths can be directly translated into tactical statements linking specific performance indicators to match outcomes.

The training of a decision tree follows a greedy, top-down procedure that selects splits based on impurity reduction measures such as Gini impurity or entropy. While decision trees are easy to interpret, they are also highly sensitive to noise and small variations in the data. In complex and noisy domains such as football event data, this sensitivity often results in high variance and limited generalisation performance. In the present study, decision trees are therefore primarily used as an interpretability benchmark and baseline model rather than as a final predictive solution.

Random Forests extend the decision tree paradigm by aggregating the predictions of multiple trees trained on different bootstrap samples of the data. Randomness is further introduced by selecting a subset of features at each split, which decorrelates individual trees and reduces variance. This ensemble strategy leads to improved stability and generalisation compared to a single tree, making Random Forests well suited to high-dimensional and noisy datasets.

In the context of football analytics, Random Forests demonstrate strong robustness and are capable of capturing complex, non-linear interactions between tactical indicators. In this thesis, Random Forests serve as a high-performance benchmark against which more interpretable models are compared. Although feature importance measures can be extracted from Random Forests, their internal decision logic remains largely opaque, limiting their usefulness for tactical explanation. This trade-off between accuracy and interpretability illustrates a recurring tension in sports analytics research.

Extreme Gradient Boosting (XGBoost) represents a further step in model complexity. XGBoost is a gradient boosting framework that builds an ensemble of decision trees sequentially, with each new tree trained to correct the residual errors of the existing ensemble. The algorithm optimises a regularised objective function that balances data fit and model complexity, making it one of the most powerful methods for structured data in many applied domains.

In this study, XGBoost achieved near-perfect performance on the training data but substantially lower performance on the validation data. This discrepancy indicates severe overfitting, reflecting the high flexibility of the model relative to the limited size of the football dataset. These findings are consistent with recent literature, which suggests that highly expressive boosting methods may not generalise well in small-sample sports analytics settings unless strong regularisation or substantially larger datasets are available. As a result, XGBoost is included in the analysis primarily as a performance upper bound rather than a practically deployable solution.

Explainable Boosting Machines (EBM) offer an effective compromise between predictive accuracy and interpretability. EBMs are an extension of Generalised Additive Models (GAMs), assuming an additive structure in which the prediction is expressed as the sum of feature-specific shape functions. Each function captures the contribution of a single feature, allowing non-linear relationships to be modelled while preserving transparency.

EBMs are trained using boosting techniques that iteratively refine these shape functions, improving accuracy without sacrificing interpretability. Optional pairwise interaction terms can be included in a controlled manner. In the context of this thesis, EBMs are particularly valuable for exploratory analysis, as feature contributions can be visualised directly and

interpreted in football-specific terms. This capability allows analysts to assess how changes in pressing intensity, spatial behaviour, or attacking indicators influence match outcomes.

The central analytical environment of the thesis is Rulex Factory, a platform specifically designed to support explainable machine learning workflows. Rulex Factory provides an integrated and modular framework for data validation, feature evaluation, model training, and performance assessment. At the core of the platform lies the Logic Learning Machine, an inherently interpretable, rule-based classifier that serves as the primary modelling technique of the study.

The Logic Learning Machine is a symbolic learning algorithm that generates models in the form of explicit IF–THEN rules. Each rule consists of a conjunction of conditions defined over intervals of input features and is associated with a predicted outcome. Unlike black-box models, the internal logic of LLM is fully transparent, allowing direct inspection and interpretation of learned decision mechanisms. This intrinsic interpretability makes LLM particularly suitable for football analytics, where model outputs must be communicable to domain experts.

The training procedure of LLM follows a structured multi-stage process. Continuous features are first discretised into meaningful intervals, transforming numerical indicators into interpretable ranges linked to tactical behaviour. Candidate logical conditions are then identified by analysing statistically significant relationships between feature intervals and the target variable. These conditions are combined into conjunctive rules that aim to balance accuracy and coverage. Finally, rule pruning and optimisation remove redundant or weak rules, ensuring compactness and robustness.(Muselli et al., n.d.)

Within this thesis, the Logic Learning Machine plays a central role. Its rule-based outputs correspond directly to tactical statements, such as combinations of passing behaviour, crossing activity, and spatial distribution associated with winning outcomes. Moreover, feature importance emerges naturally from the structure and frequency of features appearing within the learned rules, providing an intuitive and semantically meaningful measure of parameter relevance.

Rulex Factory is implemented as a visual, node-based workflow environment. Each node represents a specific analytical operation, such as feature evaluation, model training, or validation, and nodes are connected to form a transparent and reproducible pipeline. This modular architecture enforces separation between analytical stages, supports dynamic reconfiguration of experiments, and ensures that all models are trained and evaluated under identical conditions.

A key methodological component integrated into the Rulex workflow is sliding-window temporal validation. Unlike traditional cross-validation, which assumes independent and identically distributed samples, sliding-window validation respects the temporal ordering of football matches. At each iteration, models are trained on a fixed historical window and evaluated on a subsequent future window. As the window advances, older matches are discarded, and new matches are incorporated.

This validation strategy is critical in football analytics for two reasons. First, team performance evolves over time due to tactical changes, injuries, and contextual factors. Second, ignoring temporal dependencies can lead to information leakage and overly optimistic performance estimates. By embedding sliding-window validation directly into the analytical pipeline, the study ensures that performance metrics reflect realistic forecasting scenarios.

Explainable Artificial Intelligence forms the conceptual foundation of the entire technical stack. XAI refers to methods and design principles that make the behaviour and decisions of AI systems understandable to human users. In football analytics, explainability is not merely desirable but essential, as analytical insights must be assessed for plausibility and integrated with expert knowledge.

The section distinguishes between intrinsic explainability, achieved through inherently interpretable models, and post-hoc explainability, which attempts to explain black-box models after training. While post-hoc methods such as permutation importance or Shapley-based techniques are powerful, they introduce an additional layer of approximation. For this reason, the thesis prioritises intrinsically interpretable approaches wherever possible.

Feature importance plays a central role as an explainability mechanism. In this study, feature importance is assessed through ROC–AUC-based ranking, model-specific importance measures, and the frequency and structure of features appearing within LLM rules. This dual approach combines statistical relevance with semantic interpretability, ensuring that importance measures reflect both predictive contribution and tactical meaning.

In summary, this section detailed the technical stack and tools used in the thesis. By combining multiple machine learning models within the Rulex Factory platform and embedding Explainable AI principles throughout the analytical pipeline, the study provides a rigorous and interpretable framework for analysing football team structure under realistic data constraints. The emphasis on transparency, temporal validation, and rule-based knowledge extraction ensures that the resulting insights are both scientifically robust and practically meaningful.

Experimental Results and Conclusion

This section presents the experimental evaluation of the proposed football analytics framework and discusses the implications of the results in terms of predictive performance, interpretability, and methodological robustness. The experimental design, evaluation metrics, comparative results across models, and the final conclusions are all aligned with the overarching objective of the thesis: analysing football team structure and performance parameters through an explainable machine learning approach under realistic data constraints.

Train–Test Splitting Strategy

A central methodological choice in the experimental design is the adoption of a temporal train–test splitting strategy. In football analytics, match data are inherently time-dependent,

as team strength, tactical approaches, injuries, transfers, and contextual factors evolve throughout a season. Treating matches as independent and identically distributed observations, as in random cross-validation, risks introducing information leakage and inflating performance estimates.

To address this issue, the study employs sliding-window temporal validation. At each iteration, models are trained on a fixed historical window of matches and evaluated on the immediately following future window. The window then advances in time, discarding older matches and incorporating new ones. This procedure ensures that only past information is used to predict future outcomes and that model evaluation closely reflects real-world deployment conditions.

Two complementary experimental settings are considered. In the single-competition setting, models are trained and evaluated within a single league, allowing the analysis of league-specific tactical patterns under relatively homogeneous conditions. In the multi-competition setting, data from several major European leagues with identical season lengths are combined, and a competition identifier is included as a feature. This setup enables the assessment of cross-league generalisation while maintaining temporal consistency. Together, these configurations provide insight into both within-league robustness and cross-context adaptability.

Evaluation Metrics

Model performance is evaluated using a multi-metric strategy that balances interpretability and discrimination ability. Accuracy is reported as a straightforward measure of correct classification, but is not used in isolation due to its sensitivity to class imbalance and decision thresholds. Instead, accuracy is complemented by the Area Under the Receiver Operating Characteristic Curve (AUC), which measures a model's ability to discriminate between outcomes across all possible thresholds.

AUC is particularly well suited to football analytics, where match outcomes are stochastic and operational decision thresholds may vary depending on tactical objectives. For each model, both training and test metrics are reported, enabling the assessment of generalisation, overfitting, and stability across temporal splits. Large discrepancies between training and test performance are interpreted as indicators of excessive model complexity or insufficient regularisation.

Comparative Experimental Results

The experimental evaluation compares five modelling approaches: Decision Tree, Random Forest(Biau & Scornet, 2016), XGBoost(Chen & Guestrin, 2016), Explainable Boosting Machine (EBM)(Nori et al., 2021), and the Logic Learning Machine (LLM)(Muselli et al., n.d.). All models are trained on the same engineered feature space and evaluated using consistent validation protocols to ensure a fair comparison.

The results reveal a clear stratification of models. Decision Trees exhibit modest accuracy and AUC, reflecting their limited expressive capacity and high variance in noisy football data.

Random Forest models achieve strong and stable performance on both training and test sets, indicating robust generalisation and effective handling of non-linear feature interactions. XGBoost attains near-perfect performance on the training data but suffers from substantial degradation on the test data, clearly signalling severe overfitting. This behaviour confirms the risks associated with highly expressive boosting methods in small-sample, temporally structured sports datasets.

Explainable Boosting Machines occupy an intermediate position, outperforming simple decision trees while maintaining interpretability through additive feature effects. The Logic Learning Machine demonstrates competitive AUC values with a more balanced train-test behaviour than XGBoost, particularly in the single-competition setting. Across all models, zone-based passing and crossing features consistently emerge among the most influential predictors, supporting the validity of the feature engineering strategy adopted in the thesis.

Logic Learning Machine Results

Given its central role in the study, the Logic Learning Machine is analysed in greater detail. In the single-competition sliding-window evaluation, LLM achieves higher accuracy and AUC compared to the multi-competition setting, reflecting the relative homogeneity of tactical patterns within a single league. When trained across multiple competitions, performance decreases moderately, highlighting the increased difficulty of generalising across leagues with distinct playing styles. Importantly, however, the gap between training and test performance remains smaller than that observed for XGBoost, indicating greater robustness under limited and heterogeneous data.

Experiments with varying feature set sizes show that LLM maintains stable performance as the number of attributes increases, with incremental gains in accuracy and AUC. The extracted rules consistently involve zone-based passing distributions, defensive action frequencies, and crossing behaviour. These findings confirm that LLM captures structural tactical regularities rather than spurious correlations and that its rule-based outputs provide interpretable explanations aligned with football domain knowledge.

Discussion of Results

The results underscore the importance of temporally consistent evaluation. Sliding-window validation produces lower but more realistic performance estimates than random splits, reinforcing prior findings that temporal leakage can substantially inflate results in sports analytics. This methodological choice is therefore essential for credible evaluation.

The comparative analysis also illustrates the trade-off between predictive performance and interpretability. Ensemble methods such as Random Forest and XGBoost offer high accuracy but limited transparency, while interpretable models provide actionable insights at the cost of slightly lower raw performance. Among the interpretable approaches, the Logic Learning Machine stands out by achieving a favourable balance, combining competitive discriminatory power with full semantic transparency through explicit rules.

Overfitting emerges as a key concern for highly flexible models. The pronounced overfitting observed in XGBoost highlights the risks of deploying complex models in football analytics without sufficient data or strong regularisation. In contrast, LLM and EBM exhibit more controlled complexity, leading to better temporal stability and generalisation.

The consistency of top-ranked features across models, particularly zone-based passing and crossing metrics, strengthens confidence in the feature engineering pipeline. The alignment between statistical feature importance and logical importance derived from LLM rules further enhances the interpretability and credibility of the findings.

Conclusion

This thesis investigated football team structure and the importance of performance parameters through an explainable machine learning framework grounded in event-based data. By integrating structured data engineering, domain-informed feature construction, temporal validation, and interpretable modelling within a unified workflow, the study moved beyond purely predictive objectives toward actionable knowledge extraction.

The results demonstrate that explainable models, and in particular the Logic Learning Machine, can deliver competitive predictive performance while producing transparent, tactically meaningful insights. Zone-based passing distributions, crossing behaviour, and indicators of defensive organisation consistently emerged as influential factors, suggesting that spatial control and structured team behaviour are central components of successful performance.

While limitations remain most notably the exclusion of off-ball dynamics and the constrained temporal scope, the study provides a strong methodological foundation for future research. Overall, the findings support the conclusion that explainable machine learning is a viable and effective approach for football analytics, enabling models that are not only accurate but also interpretable, trustworthy, and practically relevant for decision-making in professional football.

Abstract

The growing availability of detailed football event data has fostered extensive use of machine learning techniques in performance analysis and outcome prediction. Nevertheless, many high-performing models adopted in football analytics operate as black boxes, limiting their interpretability and practical usefulness for coaches, analysts, and decision-makers. This thesis addresses this gap by investigating football team structure and the relative importance of performance parameters through an explainable machine learning framework, with a specific focus on transparency, robustness, and tactical interpretability.

The study develops a complete end-to-end analytical pipeline based exclusively on event-level football data. Raw hierarchical JSON files are transformed into a structured relational database using a reproducible data acquisition and integration process. From this database, team-level features are engineered through a domain-driven approach that prioritises interpretability over representational complexity. The resulting feature space includes aggregated statistical indicators, established tactical metrics such as Passes Per Defensive Action (PPDA) and Expected Goals (xG), and zone-based spatial features that describe how teams distribute actions across defensive, middle, and attacking thirds of the pitch.

Multiple machine learning models are evaluated under a temporally consistent sliding-window validation scheme to reflect realistic forecasting scenarios and to avoid information leakage. Comparative analysis highlights the trade-off between predictive performance and interpretability across modelling approaches. While ensemble methods achieve strong predictive accuracy, they offer limited transparency and show sensitivity to overfitting in small-sample football datasets. In contrast, the Logic Learning Machine (LLM), implemented within the Rulex platform, demonstrates competitive discriminatory performance while generating explicit IF–THEN rules that directly link combinations of tactical behaviours to match outcomes.

The results show that spatially grounded passing patterns, crossing behaviour, and indicators of pressing intensity consistently emerge as influential features across competitions. Overall, the thesis demonstrates that explainable machine learning provides a rigorous and practically meaningful framework for football analytics, enabling the extraction of transparent, actionable insights into team structure while maintaining methodological validity and reproducibility.

Chapter 1

Introduction

1.1 Background and Motivation

Over the last decade, football has undergone a profound transformation driven by the increasing availability of high-resolution event data and the rapid development of data-driven analytical methods. Professional clubs, national teams, and governing bodies now rely extensively on performance analytics to support decision-making in scouting, match preparation, tactical planning, and player development. Advances in tracking systems, event logging platforms, and data infrastructures have enabled the collection of millions of on-ball and off-ball events across competitions and seasons, providing an unprecedented opportunity to quantitatively analyse team behaviour and match dynamics.

Within this context, machine learning (ML) techniques have emerged as powerful tools for extracting patterns from complex, high-dimensional football data. Supervised learning models are commonly applied to tasks such as match outcome prediction, player valuation, expected goals (xG) estimation, and tactical profiling. However, despite their strong predictive capabilities, many state-of-the-art models, particularly ensemble and deep learning approaches, are often criticised for their lack of interpretability. In applied football analytics, where decisions must be communicated clearly to coaches, analysts, and technical staff, predictive accuracy alone is insufficient. Understanding *why* a model produces a given output, and *which factors* drive performance is equally critical.

This need has led to growing interest in Explainable Artificial Intelligence (XAI) within sports analytics. XAI methods aim to bridge the gap between predictive performance and interpretability by producing transparent models, human-readable rules, or post-hoc explanations that reveal the contribution of individual features. In football, explainability enables analysts to move beyond black-box predictions and instead identify interpretable tactical patterns, such as pressing intensity, passing structure, or spatial dominance, that are associated with winning outcomes. Recent academic literature highlights that interpretable models not only improve trust in analytics systems but also enhance their practical usability in professional sports environments (Bunker & Thabtah, 2019; Guidotti et al., 2019).

Another important challenge in football analytics concerns the heterogeneity of competitions and playing styles. Tactical behaviours and match dynamics vary significantly across leagues due to differences in tempo, physical intensity, refereeing standards, and cultural approaches to the game. As a result, analytical models trained on aggregated data risk capturing league-specific artefacts rather than generalizable football principles (Belkin et al., 2019). Temporal effects further complicate this problem, as team performance evolves throughout a season due to injuries, transfers, fatigue, and tactical adjustments. Robust analytical pipelines must therefore account for both cross-competition variability and time-dependent dynamics to ensure realistic and reliable evaluations.

In response to these challenges, the present thesis builds upon a curricular internship conducted at Rulx Innovation Labs, focusing on the application of explainable machine learning methods to football event data. The project integrates data engineering, feature

construction, and model evaluation within a unified analytical workflow, with the explicit goal of understanding team structure and the relative importance of performance parameters. The work leverages open-access football data and the Logic Learning Machine (LLM) algorithm implemented in the Rulx platform, which is designed to generate interpretable IF–THEN rules directly from data.

1.2 Football Event Data and Tactical Representation

Event-based football datasets provide a granular representation of match activity by recording every on-ball action, such as passes, shots, duels, fouls, and interceptions, together with their spatial and temporal attributes. Among the publicly available sources, the Wyscout Open Data repository has become one of the most widely used benchmarks in academic football analytics, as it covers multiple top-tier European leagues and international tournaments with consistent annotation standards.

While raw event data are rich, they are not directly suitable for machine learning applications. Effective analysis requires a careful transformation of low-level events into meaningful, higher-level features that reflect tactical concepts. Examples include aggregated passing volumes, spatial distributions of actions, pressing indicators, and chance-creation metrics. In recent years, metrics such as Passes Per Defensive Action (PPDA) and Expected Goals (xG) have become standard analytical tools for quantifying pressing intensity and chance quality, respectively. These indicators provide a bridge between data-driven modelling and football domain knowledge, making them particularly valuable for explainable analyses.

Beyond aggregate metrics, spatial context plays a crucial role in understanding team behaviour. Dividing the pitch into functional zones, such as defensive, middle, and attacking thirds, allows analysts to capture how teams distribute actions across space and how their tactical focus changes depending on match context. Zone-based features have been shown to improve both predictive performance and interpretability by linking numerical attributes to intuitive tactical notions, such as territorial control or attacking emphasis (Rein & Memmert, 2016).

The present work adopts this perspective by combining event-level data with spatial and contextual feature engineering. Raw JSON files are transformed into a relational database, enabling systematic extraction of match- and team-level statistics through structured queries. These features are then enriched with combined metrics (PPDA, xG) and zone-based indicators, forming a comprehensive representation of team performance suitable for explainable machine learning.

1.3 Explainable Machine Learning in Sports Analytics

Traditional machine learning approaches in football analytics such as Random Forests, Gradient Boosting, and neural networks, have demonstrated strong predictive performance, particularly for outcome prediction and player evaluation tasks. However, their internal decision mechanisms are often opaque, limiting their adoption in professional settings where transparency and accountability are essential. This limitation is especially relevant when analytical insights are intended to inform tactical decisions, as coaches and analysts must be able to interpret and validate model outputs.

Explainable AI addresses this issue by either designing inherently interpretable models or applying explanation techniques that make complex models understandable. Rule-based learning algorithms, such as the Logic Learning Machine, belong to the former category. They produce models in the form of logical rules that directly link conditions on input features to outcomes. Each rule can be interpreted as a tactical statement, for example, describing combinations of passing behaviour and defensive actions that increase the probability of winning.

Recent studies emphasise that explainable models are particularly well-suited to sports analytics, as they align naturally with the way tactical knowledge is communicated within teams (Ribeiro et al., 2016). Moreover, explainable approaches facilitate comparative analyses across competitions and seasons, enabling researchers to investigate whether certain tactical patterns are universally effective or context-dependent.

In this thesis, explainability is not treated as a secondary requirement but as a central design principle. Feature selection is performed using ROC–AUC-based ranking, ensuring that only the most informative attributes are passed to the model. Model evaluation is conducted through sliding-window validation, which respects the temporal structure of football seasons and mimics real-world forecasting scenarios. These methodological choices reflect best practices in modern sports analytics and are consistent with recent recommendations in the literature.

1.4 Aim of the Project

The primary aim of this thesis is to analyse football team structure and quantify the importance of performance parameters using an explainable machine learning framework.

More specifically, the objectives of the project are:

1. To design and implement a complete football analytics pipeline that transforms raw event-based JSON data into a structured, machine-learning-ready dataset through database construction, feature engineering, and spatial analysis.
2. To identify and rank the most influential tactical and performance-related features associated with match outcomes using statistically grounded evaluation metrics, with particular emphasis on interpretability.
3. To apply the Logic Learning Machine (LLM) algorithm within the Rulex platform in order to extract transparent, rule-based models that describe interpretable relationships between team behaviours and winning outcomes.
4. To evaluate model performance under realistic temporal conditions using sliding-window validation across multiple top European leagues, assessing both predictive stability and generalisation.
5. To investigate cross-competition differences in tactical patterns, and analyse whether certain features or rules are league-specific or consistent across different football contexts.

Chapter 2

Data Acquisition and Integration

2.1 Overview

Data acquisition and integration represent a foundational phase in football analytics pipelines, particularly when machine learning methods are employed to study team structure and performance determinants. The predictive and explanatory quality of any model is directly dependent on the reliability, granularity, and structural coherence of the underlying data. In the football domain, this challenge is amplified by the heterogeneity of event-based datasets, the hierarchical organisation of competitions, teams, matches, and players, and the high temporal and spatial resolution required to capture tactical behaviour.

This chapter describes the data acquisition and integration strategy adopted in this thesis. Following established best practices in sports analytics, the workflow is designed to transform raw football event data into a structured, machine learning ready representation that supports both predictive modelling and explainability. The chapter is organised into three main sections: (i) the data sources used and their analytical relevance, (ii) the parsing and restructuring of hierarchical JSON data, and (iii) the construction of a relational SQLite database that serves as the backbone of the analytical pipeline.

2.2 Data Sources

2.2.1 Football Event Data in Modern Analytics

Event-based football data constitute one of the most widely used data modalities in contemporary sports analytics. Unlike tracking data, which records continuous player trajectories, event data logs discrete on-ball actions together with contextual information such as time, location, player identity, and action type. This representation offers a favourable trade-off between interpretability, availability, and computational complexity, making it particularly suitable for large-scale analytical studies and explainable machine learning applications (Rein & Memmert, 2016).

Recent research highlights that event data are especially effective for modelling team-level tactical behaviour, as they enable the aggregation of actions into meaningful performance indicators such as passing structure, pressing intensity, and attacking (Power et al., 2017; Goes et al., 2021). Moreover, event-based features are more easily translated into human-understandable concepts, which align naturally with the objectives of explainable artificial intelligence in sports.

2.2.2 Wyscout Open Data Repository

The primary data source used in this thesis is the **Wyscout Open Data** repository, a publicly available dataset that has become a reference standard in football analytics research. The dataset provides detailed event-level information for multiple top-tier European leagues—including the English Premier League, Italian Serie A, Spanish La Liga, German Bundesliga, and French Ligue 1, as well as major international tournaments such as the FIFA World Cup and UEFA European Championship.

Each competition is distributed as a collection of structured JSON files, covering:

- **Matches:** match identifiers, dates, teams involved, and final outcomes;
- **Teams:** team identifiers and competition affiliations;

- **Players:** player metadata and team associations;
- **Events:** detailed logs of all on-ball actions (passes, shots, duels, fouls, interceptions, etc.), including spatial coordinates and temporal markers.

The choice of this dataset is motivated by several factors. First, its multi-competition coverage enables cross-league analyses and comparative studies of tactical behaviour. Second, the high granularity of event annotations supports advanced feature engineering, including spatial and temporal aggregations. Third, the dataset is widely used in peer-reviewed research, which facilitates methodological comparability and reproducibility (Decroos et al., 2019; Andrienko et al., 2007).

2.2.3 Data Selection Rationale in the Project

In the present project, the Wyscout dataset was selected to support the analysis of team structure and parameter importance using machine learning. Multiple competitions were integrated into a unified analytical framework to ensure sufficient variability in tactical styles and match contexts.

Particular emphasis was placed on competitions with complete and homogeneous season structures. This choice later enabled the implementation of temporally consistent validation strategies, such as sliding-window evaluation, which are essential for realistic performance assessment in sports forecasting scenarios.

2.3 JSON Parsing and Structure

2.3.1 Hierarchical Nature of Football JSON Data

JSON (JavaScript Object Notation) is a flexible, human-readable format commonly used to store hierarchical and semi-structured data. While its nested structure is well suited for representing complex relationships such as events belonging to matches, which in turn belong to competitions, it poses significant challenges for statistical analysis and machine learning. Most analytical methods require tabular or relational representations, where entities and their attributes are explicitly defined and linked via keys.

Football event JSON files typically contain deeply nested objects. For example, an individual event may include nested fields for player identity, team identity, event type, sub-event type, spatial coordinates, and qualifiers. Figure 2.1 illustrates the hierarchical schema of the JSON files used in this project, highlighting the nested structure linking competitions, matches, teams, players, and event-level attributes. Without systematic parsing, these structures remain inaccessible to standard data analysis tools (Gudmundsson & Horton, 2017; Andrienko et al., 2007).

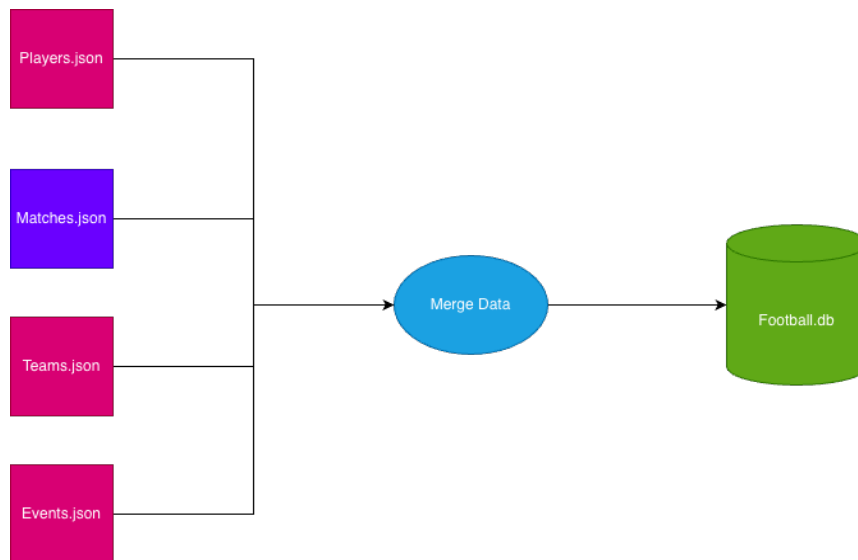


Figure 2.1 Simplified schema of the hierarchical JSON structure used in the Wyscout football event dataset, showing the relationships between competitions, matches, teams, players, and event attributes.

2.3.2 Parsing Strategy and Tools

To address these challenges, a dedicated parsing workflow was implemented using Python. The approach follows recent recommendations in sports data engineering, which advocate for early normalisation of raw data into analytically meaningful entities (Pappalardo et al., 2019)

The parsing process relied on the following components:

- JSON parsing libraries to read and navigate nested structures;
- Data manipulation tools to flatten hierarchical objects into structured tables;
- Explicit key management to preserve relationships between matches, teams, players, and events.

Each JSON file was processed independently, extracting relevant fields and resolving nested attributes into atomic variables. For instance, event-level data were decomposed into explicit columns describing action type, outcome, spatial coordinates, and temporal information, while retaining references to match and team identifiers.

2.3.3 Relational Decomposition of Events

A key design decision in the parsing stage was the relational decomposition of event data. Rather than maintaining a single wide table containing all information, the data were organised into logically distinct entities, such as matches, teams, players, and events. This approach reflects principles of database normalisation and offers several advantages:

1. **Consistency:** shared information (e.g., team metadata) is stored once and referenced via identifiers;
2. **Scalability:** new competitions or seasons can be added without restructuring existing data.

3. **Analytical flexibility:** event-level data can be aggregated or filtered efficiently through relational queries.

This decomposition mirrors the conceptual model commonly adopted in professional football analytics platforms.

2.4 Database Construction with SQLite

2.4.1 Rationale for a Relational Database Approach

Once parsed, the football data must be stored in a format that supports efficient querying, aggregation, and feature extraction. Relational databases are particularly well suited for this purpose, as they provide structured schemas, enforce data integrity, and enable complex analytical queries using SQL.

Recent studies in sports analytics emphasise that relational databases offer significant advantages over flat files when dealing with multi-entity, multi-season football datasets (Herold et al., 2019). They allow analysts to dynamically generate features at different granularities (event, match, team, season) while maintaining reproducibility and traceability.

2.4.2 Choice of SQLite

In this project, SQLite was selected as the database management system. SQLite is a lightweight, file-based relational database engine that requires no server configuration and integrates seamlessly with Python-based data science workflows. Despite its simplicity, SQLite supports standard SQL operations and is capable of handling datasets of the size typically encountered in academic football analytics.

In the HydrocamCollect workflow, local file and upload metadata are persisted in an embedded relational database (SQLite) to support reliable operation on resource-constrained edge devices. The SQLite database is maintained as a local file within the project directory and is accessed through dedicated read/write routines to track the status of captured imagery and subsequent cloud transfers. SQLite was selected because its serverless and lightweight architecture reduces deployment overhead (no separate database service is required), while still providing transactional integrity for frequent metadata updates during unattended data acquisition on platforms such as the Raspberry Pi.(Neupane et al., 2026)

2.4.3 Database Schema Design

The database schema was designed to reflect the logical structure of football competitions and events. Core tables included:

- **Matches table**, containing match-level metadata and outcomes.
- **Teams table**, storing team identifiers and competition affiliations.
- **Players table**, describing player identities and team associations.
- **Events table**, recording all on-ball actions with references to matches, teams, and players.

Primary keys were assigned to each table, and foreign keys were used to enforce referential integrity across entities. The resulting relational structure is summarised in the Entity–Relationship Diagram (ERD) shown in Figure 2.2, which provides a high-level overview of the database schema and its key relationships. This design ensures that every event can be traced back to its corresponding match and team, enabling consistent aggregation and analysis.

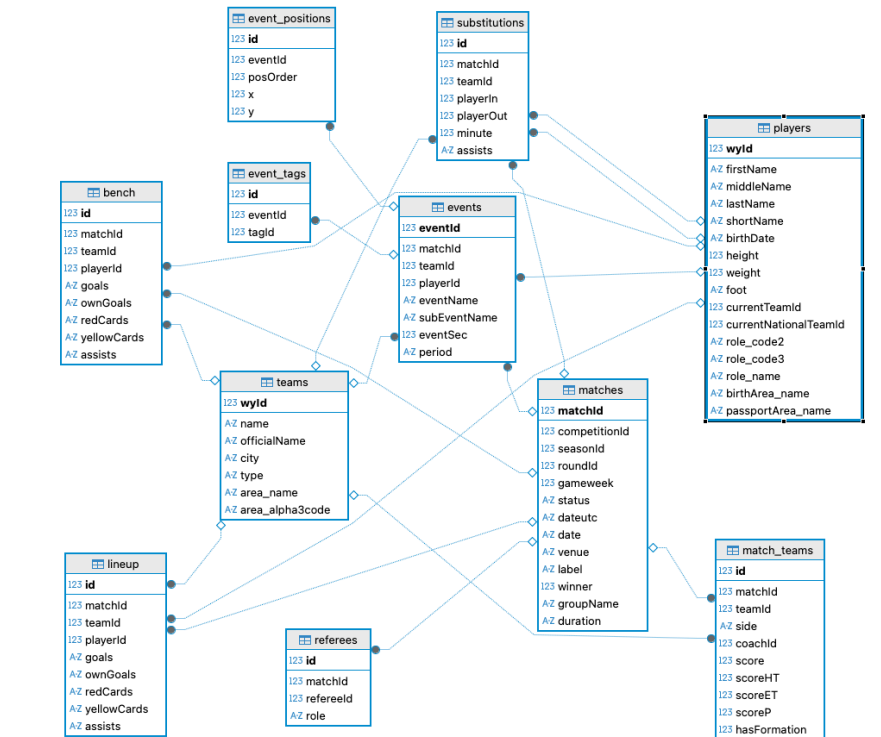


Figure 2.2 Entity Relationship Diagram (ERD) of the SQLite database used in the project, illustrating the main tables (matches, teams, players, events) and their primary–foreign key relationships.

2.4.4 Integration and Feature Readiness

Once populated, the SQLite database served as the central integration layer of the project. SQL queries were used to compute match- and team-level statistics, aggregate event counts, and derive advanced indicators that later fed into the machine learning pipeline. By centralising data access within the database, feature extraction became both transparent and reproducible.

This database-driven approach enabled the systematic computation of tactical indicators and ensured compatibility with downstream analytical tools, including the Rulex platform.

2.5 Summary

This chapter has presented the data acquisition and integration methodology underpinning the thesis. Starting from open-access football event data, a structured pipeline was developed to parse hierarchical JSON files, normalise their contents, and integrate them into a relational SQLite database. These steps are essential prerequisites for the explainable machine learning analyses presented in subsequent chapters.

By adhering to established best practices in sports data engineering and aligning with recent academic literature, the proposed workflow ensures data quality, analytical flexibility, and methodological transparency. Most importantly, it provides a robust foundation for investigating team structure and parameter importance through interpretable machine learning models.

Chapter 3

Feature Engineering Pipeline

3.1 Overview and Methodological Motivation

Feature engineering represents a critical step in football analytics pipelines, particularly when the objective is to analyse team structure and quantify the importance of performance parameters using machine learning. In contrast to domains where massive, clean datasets are available, football analytics often operates under data scarcity, heterogeneous contexts, and high observational noise. These characteristics significantly constrain the applicability of data-hungry models such as deep neural networks, which typically require very large training samples and stable feature representations to generalise effectively.

Recent studies in sports analytics emphasise that, in small to medium sample regimes, carefully designed, domain informed features often outperform automatically learned representations, both in predictive accuracy and interpretability (Herold et al., 2019; Rein & Memmert, 2016). Moreover, football event data are inherently noisy: not all recorded actions carry equal tactical relevance, and many low-level events are weakly informative when considered in isolation. As a result, relying directly on raw event counts or deep learning-based feature extraction risks amplifying noise rather than uncovering meaningful patterns. For these reasons, the present project adopts a feature-centric modelling strategy. Instead of attempting to learn latent representations through deep architectures, the pipeline focuses on transforming raw event data into structured, aggregated, and interpretable features that encode tactical information at the team level. This approach aligns naturally with the explainable machine learning framework adopted in the Rulx platform and supports the extraction of transparent rules linking team behaviour to match outcomes.

This chapter describes the feature engineering pipeline in three stages: (i) SQL-based statistical features, (ii) tactical indicators such as PPDA and xG, and (iii) zone-based spatial features. Together, these components form a compact yet expressive representation of team performance suitable for explainable modelling.

3.2 SQL-Based Statistical Features

3.2.1 Rationale for Aggregated Statistical Features

The first level of feature engineering involves the extraction of aggregated statistical features from the relational database constructed in Chapter 2. While individual events provide fine-grained information, machine learning models benefit from representations that summarise behaviour over meaningful temporal units, such as matches. Aggregation reduces variance,

mitigates annotation noise, and produces features that are more stable across games and competitions.

From a methodological standpoint, this aggregation step is particularly important given the limited size of the training dataset. With only a finite number of matches per competition and season, directly modelling event sequences would result in an unfavourable ratio between feature dimensionality and sample size. Aggregated statistics therefore act as a form of implicit regularisation, reducing model complexity while preserving salient information (Goes et al., 2021).

3.2.2 SQL-Based Feature Extraction

Using the SQLite database as a central integration layer, SQL queries were employed to compute match-level and team-level statistics. These features include, but are not limited to:

- total number of passes, shots, crosses, duels, and fouls;
- breakdowns of pass types (simple, high, smart);
- counts of defensive actions such as tackles and interceptions;
- Home and away team statistics computed symmetrically.

The use of SQL offers several advantages. First, it ensures reproducibility, as each feature is defined by an explicit query. Second, it allows efficient aggregation across large event tables without duplicating data in memory. Third, it enforces a clear separation between data storage and feature computation, which improves transparency and auditability.

These SQL-based features form the backbone of the analytical dataset and provide a baseline description of team activity before introducing more advanced tactical indicators.

3.2.3 Limitations of Raw Statistical Features

Despite their usefulness, purely statistical features suffer from important limitations. Event counts alone do not capture context, quality, or intent. For example, a high number of passes may reflect either effective ball circulation or sterile possession with little offensive impact. Similarly, defensive actions are not inherently positive, as they may indicate that a team is under sustained pressure.

These limitations motivate the introduction of composite tactical indicators, which combine multiple event types into higher-level metrics grounded in football theory. Such indicators are essential to move beyond descriptive statistics and towards explanatory analysis.

3.3 Tactical Indicators: PPDA and xG

3.3.1 Motivation for Tactical Indicators

Tactical indicators play a crucial role in modern football analytics by translating raw event data into interpretable measures of team behaviour. Unlike deep learning based feature extraction, which attempts to infer representations automatically, tactical indicators are explicitly designed to reflect known football concepts. This design choice is particularly

advantageous when data are limited and noisy, as it embeds domain knowledge directly into the feature space.

Among the many proposed indicators, Passes Per Defensive Action (PPDA) and Expected Goals (xG) have emerged as two of the most widely adopted and empirically validated metrics in both academic research and professional practice (Power et al., 2017; Bransen & Van Haaren, 2019)

3.3.2 Passes Per Defensive Action (PPDA)

PPDA is a measure of pressing intensity, defined as the number of passes an opponent is allowed to complete before the defending team performs a defensive action. Lower PPDA values correspond to more aggressive pressing, while higher values indicate a more passive defensive approach.

In this project, PPDA was computed using SQL queries that aggregated opponent passes and defensive actions at the match level. The metric captures an important structural aspect of team behaviour: how actively a team attempts to disrupt the opponent's build-up play.

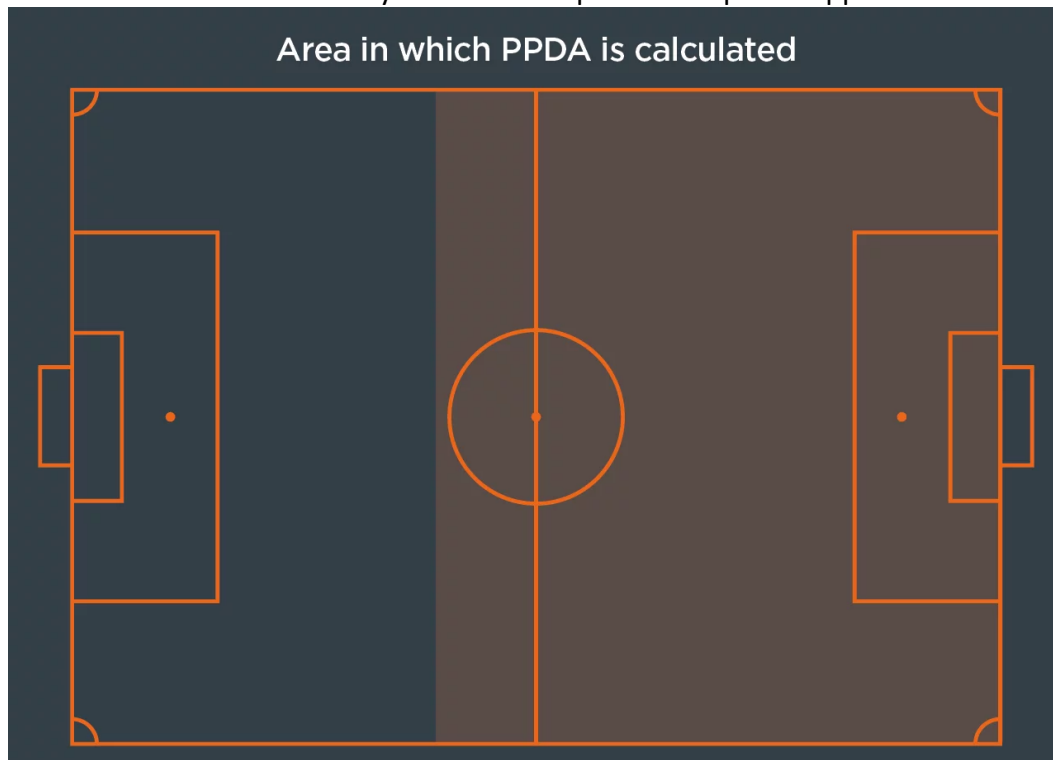


Figure 3.1. Conceptual illustration of the PPDA metric, showing opponent passes and defensive actions used to quantify pressing intensity

PPDA is particularly suitable for explainable modelling, as it corresponds to a clear tactical concept that can be directly interpreted by coaches and analysts. Moreover, it has been shown to correlate with match outcomes and territorial dominance across leagues (Link & Hoernig, 2017)

3.3.3 Expected Goals (xG)

Expected Goals (xG) quantify the quality of goal-scoring opportunities by estimating the probability that a given shot results in a goal. Unlike raw shot counts, xG accounts for

contextual factors such as shot distance, angle, and situation, thereby providing a more nuanced measure of attacking effectiveness.

In the present project, xG values were assigned to shot events using their spatial and contextual attributes and then aggregated at the team level. This approach allows the model to distinguish between teams that create many low-quality chances and those that generate fewer but more dangerous opportunities.

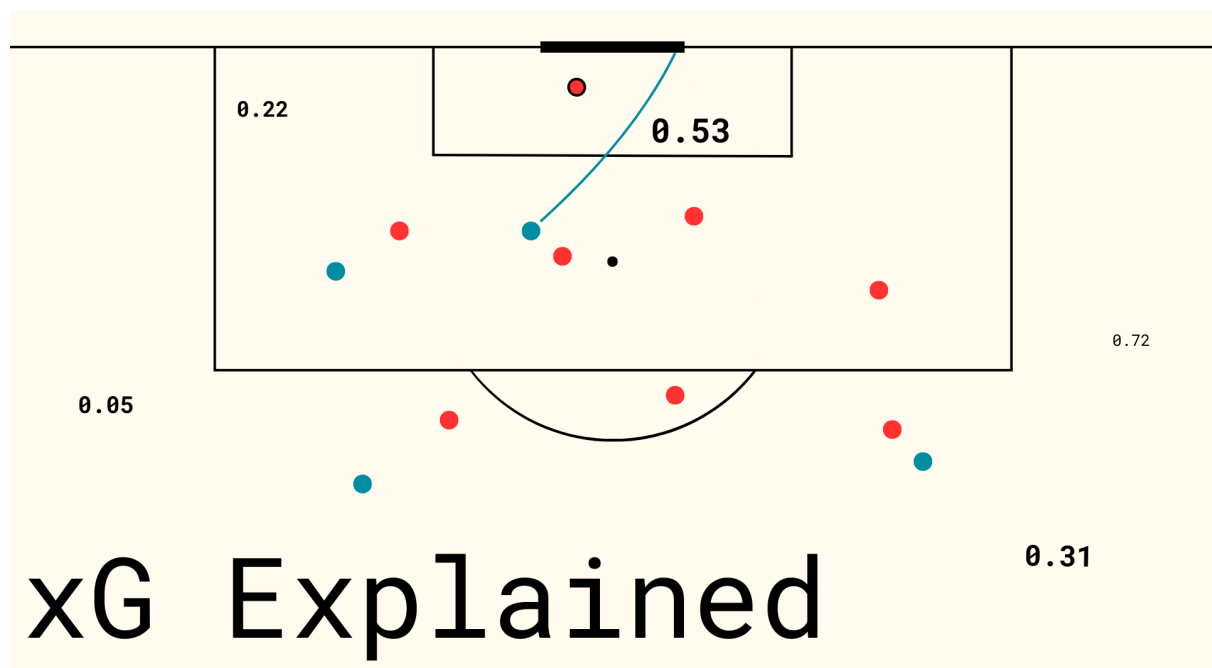


Figure 3.2. Example of shot locations and corresponding xG values, illustrating how chance quality varies across the pitch.

From a methodological perspective, xG serves as a noise-reducing feature, as it smooths out the randomness inherent in goal scoring. This property is especially valuable in small-sample settings, where outcomes such as goals and wins are subject to high variance (Lucey et al., n.d.).

3.4 Zone-Based Features

3.4.1 Motivation for Spatial Abstraction

While tactical indicators capture qualitative aspects of play, they do not fully describe where actions occur on the pitch. Spatial context is essential for understanding team structure, as identical actions can have very different tactical meanings depending on their location.

Directly modelling continuous spatial coordinates, however, would dramatically increase feature dimensionality and sensitivity to noise. To address this issue, the project adopts a zone-based spatial abstraction, which discretises the pitch into meaningful regions and aggregates events accordingly. This approach balances expressiveness and robustness and is well supported in recent football analytics literature (Fernandez et al., n.d.; Goes et al., 2021).

3.4.2 Pitch Division and Feature Construction

The football pitch was divided into three main zones:

- **Defensive third (DEF)**

- **Middle third (MID)**
- **Attacking third (ATT)**

Using event coordinates, each action was assigned to one of these zones. Zone-specific features were then computed, such as the number of high passes in the attacking third or crosses from the defensive zone. These features capture how teams distribute their actions spatially and provide insight into tactical orientation and risk-taking behaviour.

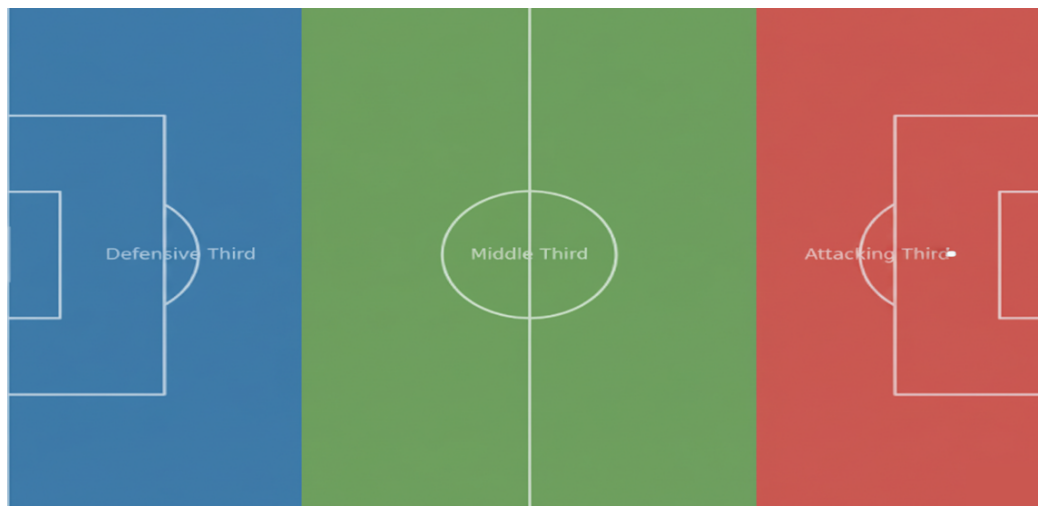


Figure 3.3. Zone-based division of the football pitch into defensive, middle, and attacking thirds is used for spatial feature engineering.

3.4.3 Benefits for Explainable Machine Learning

Zone-based features are particularly well suited for explainable machine learning. They reduce noise by aggregating spatial information and yield attributes that are immediately interpretable. For example, a rule involving “high passes in the attacking third” can be directly translated into a tactical interpretation, unlike abstract latent features produced by deep models.

Moreover, these features proved highly informative in the Rulex analysis, consistently appearing among the top-ranked attributes across multiple competitions. This empirical evidence supports the methodological choice to prioritise structured, domain-driven feature engineering over purely data-driven representation learning.

3.5 Summary

This chapter has presented the feature engineering pipeline adopted in the thesis, motivated by the constraints of limited sample size, noisy event data, and the need for interpretability. By combining SQL-based statistical features, tactical indicators such as PPDA and xG, and zone-based spatial abstractions, the pipeline constructs a compact yet expressive representation of team performance.

Rather than relying on deep learning methods ill-suited to the data regime, the approach embeds football knowledge directly into the feature space. This design choice enables robust, explainable machine learning analyses and lays the foundation for the modelling and evaluation stages presented in the subsequent chapter.

Chapter 4

Technical Stack and Tools

4.1 Introduction and Methodological Positioning

This chapter presents the complete technical stack employed in this thesis, detailing the machine learning models, analytical workflows, and explainability mechanisms used to analyse football team structure and the relative importance of performance parameters. Unlike purely predictive studies, the primary objective of this work is knowledge extraction rather than maximisation of raw accuracy. Consequently, the selection of tools and models prioritises interpretability, robustness under limited data conditions, and methodological transparency.

Football analytics presents a unique modelling environment characterised by three fundamental constraints. First, the effective sample size is intrinsically limited: even a full domestic league season yields fewer than 400 matches, and once temporal validation is imposed, the usable training data per iteration becomes substantially smaller. Second, football event data are structurally noisy, as recorded actions vary widely in tactical relevance and are influenced by contextual factors such as match state, opposition strength, and refereeing. Third, applied football analytics requires human interpretability, as insights must be communicated to analysts, coaches, and decision-makers who operate in a domain governed by tactical reasoning rather than abstract statistical optimisation.

These constraints render deep learning approaches unsuitable for the present study. Deep neural networks typically require orders of magnitude more data to learn stable representations and, even when successful, produce latent features that are difficult to interpret in tactical terms ((Herold et al., 2019). Instead, this thesis adopts a model-centric and explainability-driven approach, combining classical machine learning, ensemble methods, and inherently interpretable algorithms within a unified analytical framework.

The technical stack is structured around three pillars:

1. Machine learning models are used for prediction and feature evaluation.
2. Rulex Factory, which orchestrates feature selection, training, and validation.
3. Explainable Artificial Intelligence (XAI) principles, which guide model choice and interpretation.

4.2 Machine Learning Models

4.2.1 Overview of Model Selection

The model selection was deliberately designed to span multiple modelling paradigms, ranging from simple, inherently interpretable models to more complex ensemble-based approaches.

This diversity enables a systematic comparison between predictive performance, robustness under limited data, and explainability, which are central objectives of the present study.

The selected models are:

- Decision Tree
- Random Forest
- Extreme Gradient Boosting (XGBoost)
- Explainable Boosting Machine (EBM)
- Logic Learning Machine (LLM)

All models were trained on the same engineered feature space, derived from football event data and tactical indicators, and evaluated using identical temporal validation schemes based on sliding windows. This design ensures methodological consistency and prevents confounding effects arising from differences in data preprocessing or evaluation strategy.

The choice to exclude deep learning models is intentional. As discussed in previous chapters, football analytics is characterised by relatively small sample sizes, strong temporal dependencies, and noisy observational data. Under such conditions, deep neural networks are prone to overfitting and produce representations that are difficult to interpret in tactical terms (Herold et al., 2019). Instead, the selected models are well suited to structured, tabular data and allow varying degrees of interpretability, making them appropriate for analysing team structure and parameter importance.

The following subsections describe each model in detail, focusing on its learning mechanism, theoretical properties, and relevance to football analytics.

4.2.2 Decision Tree

4.2.2.1 Model Definition

A Decision Tree is a non-parametric supervised learning algorithm that recursively partitions the feature space into a hierarchical structure of decision rules. Each internal node represents a conditional test on a feature, each branch corresponds to the outcome of that test, and each leaf node assigns a class label or prediction. The resulting structure can be interpreted as a set of nested IF–THEN rules.

In football analytics, decision trees are particularly appealing because each decision path can be directly interpreted as a tactical rule, linking specific performance indicators such as passing behaviour, defensive actions, or spatial features, to match outcomes. This interpretability makes decision trees a natural baseline for explainable modelling in sports contexts (Rein & Memmert, 2016)

4.2.2.2 Training Algorithm

The training of a decision tree follows a greedy, top-down recursive partitioning procedure:

1. At the root node, the algorithm evaluates all candidate input features and all possible split thresholds.
2. For each candidate split, an impurity measure is computed, typically Gini impurity or entropy, to quantify class heterogeneity.
3. The split that maximally reduces impurity is selected.
4. The dataset is partitioned into child subsets according to the selected split.
5. Steps 1–4 are repeated recursively for each child node until a stopping criterion is met.

Common stopping criteria include a maximum tree depth, a minimum number of samples per node, or zero impurity.

Formally, for a node containing a dataset D , the optimal split s^* is defined as:

$$s^* = \operatorname{arimax}_s \left(I(D) - \sum \frac{D_k}{D} I(D_k) \right),$$

where $I(\cdot)$ denotes the impurity measure and D_k are the subsets induced by split s .

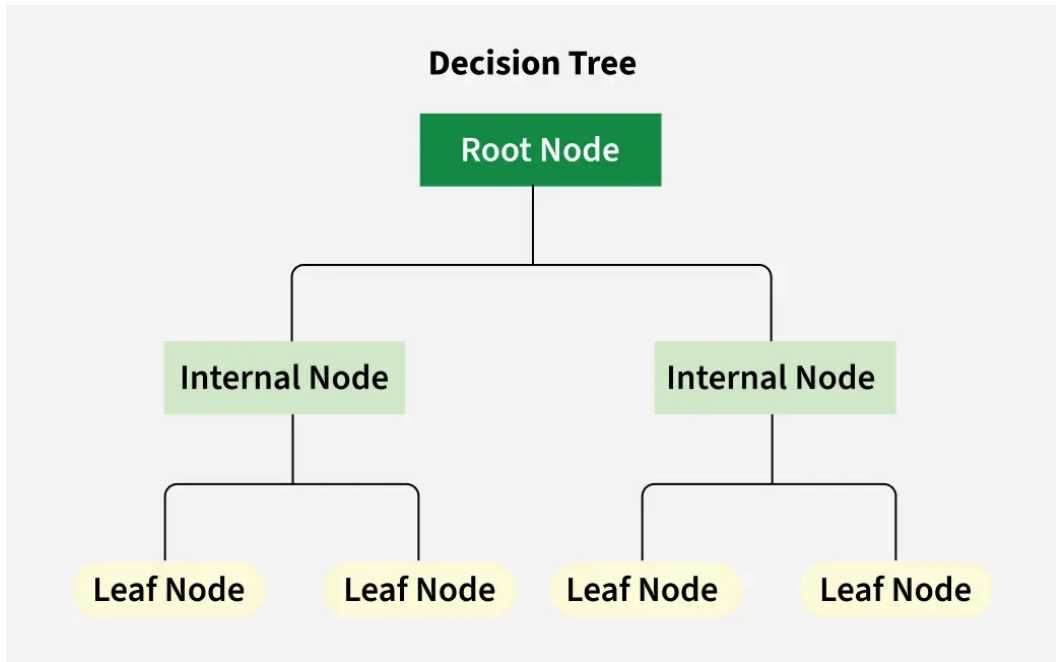


Figure 4.1. Structure of a Decision Tree.

4.2.2.3 Role in This Project

In this thesis, the Decision Tree model serves as a baseline interpretable classifier, providing a reference point for rule-based reasoning. Its explicit structure allows direct inspection of feature interactions and decision logic.

However, due to their high variance and sensitivity to noise, decision trees often exhibit limited generalisation performance in complex and noisy domains such as football event data. This behaviour is confirmed by the empirical results of the study and aligns with findings in the literature (Biau & Scornet, 2016). As a result, decision trees are primarily used for interpretability benchmarking rather than as final predictive models.

4.2.3 Random Forest

4.2.3.1 Model Definition

Random Forest is an ensemble learning method that improves upon single decision trees by aggregating the predictions of multiple trees trained on different bootstrap samples of the data. In addition to bootstrap aggregation, randomness is introduced by selecting a random subset of features at each split, which decorrelates individual trees and reduces variance.

Random Forests are particularly well suited to high-dimensional, noisy datasets and are widely used in applied machine learning tasks involving structured data (Biau & Scornet, 2016).

4.2.3.2 Training Algorithm

The training process of a Random Forest consists of the following steps:

1. Draw B bootstrap samples from the original training dataset.
2. Train one decision tree on each bootstrap sample.
3. At each node split, select a random subset of features and determine the best split only among those features.
4. Aggregate predictions from all trees using majority voting for classification tasks.

This procedure approximates the expected prediction of an ensemble of decorrelated trees, leading to improved stability and generalisation compared to a single tree.

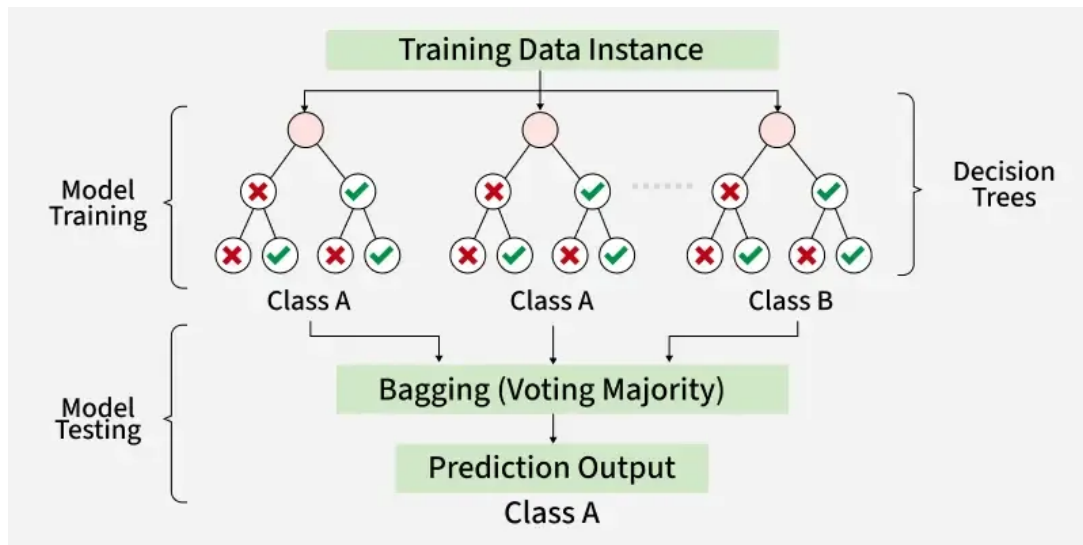


Figure 4.2. Random Forest ensemble architecture

4.2.3.3 Application to Football Data

In the context of football analytics, Random Forests demonstrate strong robustness to noise and complex, non-linear interactions between features. They are capable of capturing relationships between tactical indicators that may not be apparent in simpler models.

In this thesis, Random Forest serves as a high-performance benchmark against which more interpretable models are compared. While feature importance measures can be extracted from Random Forests, their internal decision logic remains largely opaque, limiting their usefulness for tactical explanation. This trade-off between accuracy and interpretability is a recurring theme in sports analytics research (Herold et al., 2019).

4.2.4 Extreme Gradient Boosting (XGBoost)

4.2.4.1 Model Definition

XGBoost is a gradient boosting framework that builds an ensemble of decision trees in a sequential manner. Each new tree is trained to correct the residual errors of the existing ensemble, optimising a regularised objective function that balances data fit and model complexity.

XGBoost has become one of the most popular algorithms for structured data due to its computational efficiency and strong predictive performance (Chen & Guestrin, 2016; recent applications in sports analytics reviewed in (Bransen & Van Haaren, 2019)

4.2.4.2 Training Algorithm

The XGBoost algorithm minimises the following objective function:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_t \Omega(f_t)$$

where $l(y_i, \hat{y}_i)$ is a differentiable loss function and $\Omega(f_t)$ is a regularisation term penalising the complexity of each tree f_t .

Training proceeds iteratively:

1. Initialise predictions with a constant value.
2. Compute first- and second-order gradients of the loss function.
3. Fit a new tree to the gradients.
4. Update predictions by adding the weighted contribution of the new tree.
5. Repeat until convergence or a predefined number of iterations is reached.

4.2.4.3 Observed Behaviour in This Study

In the present project, XGBoost achieved near-perfect performance on the training data but substantially lower performance on the validation data. This discrepancy indicates severe overfitting, reflecting the high flexibility of the model relative to the limited size of the football dataset.

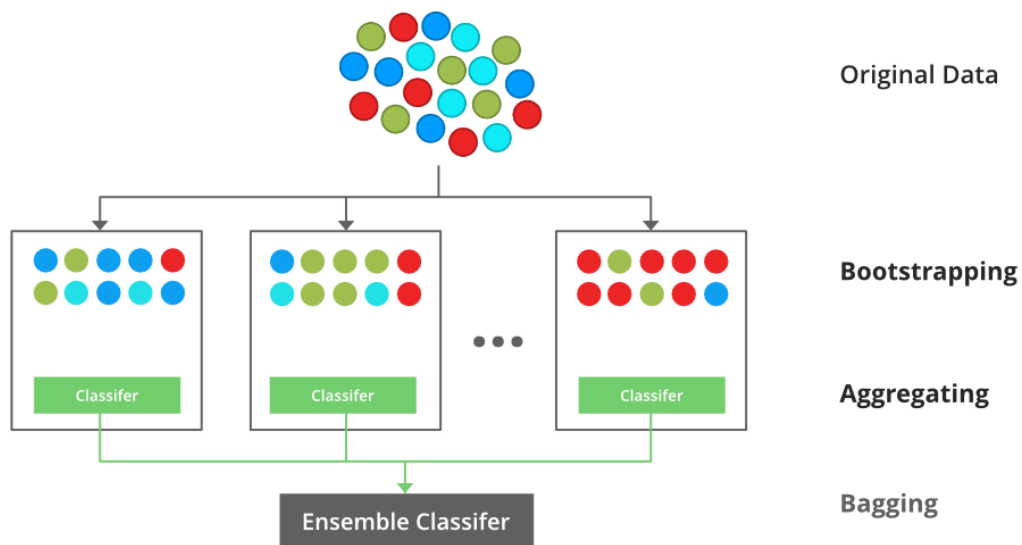


Figure 4.3. Gradient boosting process in XGBoost

This empirical outcome reinforces recent findings that highly expressive boosting methods may not generalise well in small-sample sports analytics settings unless strong regularisation and extensive data are available (Herold et al., 2019).

4.2.5 Explainable Boosting Machine (EBM)

4.2.5.1 Model Definition

Explainable Boosting Machines are an extension of Generalised Additive Models (GAMs) enhanced with boosting techniques. The model assumes an additive structure of the form:

$$g(E[y]) = \beta_0 + \sum_j f_j(x_j)$$

where each $f_j(x_j)$ is a learned shape function representing the contribution of the feature x_j to the prediction.

This structure preserves interpretability while allowing non-linear relationships between features and outcomes.

4.2.5.2 Training Algorithm

EBMs are trained by fitting shallow decision trees to learn each shape function independently. Boosting is used to iteratively refine these functions, improving predictive accuracy while maintaining the additive structure. Optional pairwise interaction terms can be included without compromising interpretability (Nori et al., 2019)

4.2.5.3 Relevance to This Project

EBM offers an effective compromise between accuracy and transparency. Feature contributions can be visualised directly, enabling analysts to assess how changes in specific football indicators influence match outcomes. This property makes EBMs particularly attractive for exploratory and explanatory analysis in football analytics.

4.3 Rulex Factory

Rulex Factory constitutes the central analytical environment used in this thesis, providing an integrated and modular framework for data processing, feature evaluation, model training, and validation. The platform is specifically designed to support explainable machine learning workflows and is therefore particularly well suited to the objectives of this study, which focus on analysing football team structure and assessing the importance of performance parameters in an interpretable manner.

At the core of Rulex Factory lies the Logic Learning Machine (LLM), an inherently interpretable rule-based classifier that serves as the primary modelling technique of the project. Rulex Factory orchestrates the entire modelling pipeline, ensuring methodological consistency across feature selection, training, and evaluation stages, while enabling direct extraction of human-readable knowledge from data.

4.3.1 Logic Learning Machine (LLM)

Model Definition

The Logic Learning Machine is a symbolic, rule-based classification algorithm that generates models in the form of explicit IF–THEN rules. Each rule represents a conjunction of conditions defined over intervals of input features and is associated with a predicted outcome. Unlike black-box models, the internal logic of LLM is fully transparent, allowing users to directly inspect and interpret the learned decision mechanisms.

From a conceptual standpoint, LLM belongs to the family of intrinsically explainable models, where interpretability is embedded in the model structure itself rather than achieved through post-hoc approximation techniques. This property is particularly valuable in football analytics, where model outputs must be interpretable in tactical terms and communicable to domain experts such as analysts and coaches (Guidotti et al., 2019)

Training Algorithm

The training procedure of the Logic Learning Machine follows a structured, multi-stage process designed to balance predictive accuracy and interpretability:

- 1. Discretisation of continuous features**

Continuous input variables are transformed into a finite set of meaningful intervals using data-driven discretisation techniques. This step is essential for football analytics, as it converts numerical performance indicators (e.g., pass counts, percentages, spatial frequencies) into interpretable ranges that can be directly linked to tactical behaviour.

- 2. Identification of candidate conditions**

Candidate logical conditions are generated by identifying feature interval pairs that exhibit statistically significant correlations with the target variable. This stage ensures that only informative conditions are considered for rule construction.

- 3. Construction of rule sets**

Individual conditions are combined into conjunctive IF–THEN rules. The algorithm seeks to maximise both accuracy (correct classification of outcomes) and coverage (the proportion of samples explained by the rules), thereby avoiding overly specific or excessively generic rules.

4. Rule pruning and optimisation

Redundant, conflicting, or weak rules are removed through pruning procedures. This final step reduces overfitting and ensures that the resulting rule set remains compact and interpretable.

This learning strategy reflects modern advances in symbolic machine learning and has been shown to produce stable and semantically meaningful models even in small-sample, noisy domains (Muselli, 2021).

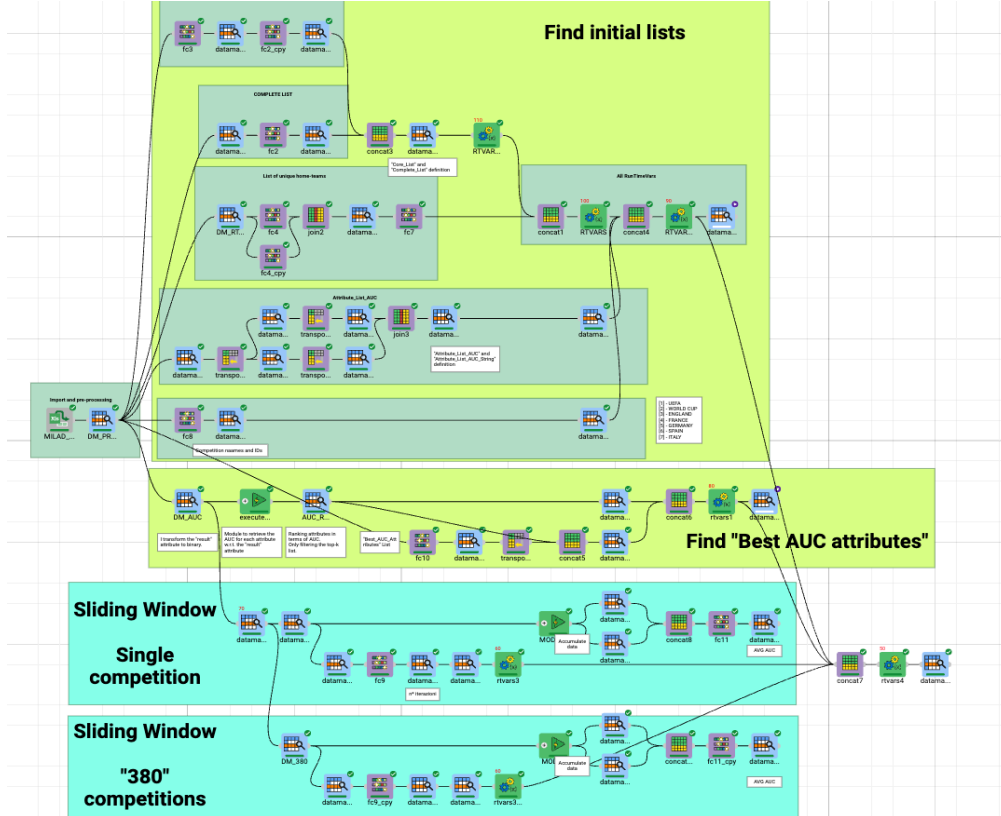


Figure 4.4. Rulex Factory visual workflow environment.

Central Role in This Thesis

The Logic Learning Machine represents the core analytical model of this thesis. Its rule-based outputs correspond directly to tactical statements, such as combinations of passing behaviour, crossing activity, and spatial distribution of actions associated with match outcomes. As a result, LLM enables the translation of statistical patterns into football-relevant knowledge.

Furthermore, the frequency and structure of features appearing within the learned rules provide a natural and intuitive measure of feature importance, complementing statistical ranking methods used earlier in the pipeline. This dual interpretability quantitative and logical makes LLM uniquely suited for explainable football analytics and aligns with the broader objectives of Explainable Artificial Intelligence (Longo et al., 2025).

4.3.2 Architecture and Workflow Design of Rulex Factory

Rulex Factory is implemented as a visual, node-based workflow environment, where each node represents a specific analytical operation, such as data validation, feature evaluation, model training, or performance assessment. Nodes are connected sequentially to form a reproducible and transparent pipeline.

This modular architecture offers several methodological advantages. First, it enforces a clear separation between different analytical stages, improving reproducibility and auditability. Second, it allows dynamic reconfiguration of workflows, enabling experiments across multiple competitions or feature subsets without manual intervention. Third, it ensures that all models are trained and evaluated under identical conditions, which is essential for fair comparison.

Within the context of this thesis, Rulex Factory integrates:

- feature evaluation through ROC–AUC analysis,
- model training using multiple algorithms,
- extraction of interpretable rules and performance metrics, and
- aggregation of results across temporal validation folds.

4.3.3 Sliding-Window Temporal Validation

A key methodological component implemented within Rulex Factory is sliding-window temporal validation. Unlike traditional cross-validation, which assumes independent and identically distributed samples, sliding-window validation respects the temporal ordering of football matches.

At each iteration, the model is trained using data from a fixed historical window and evaluated on a subsequent future window. As the window slides forward in time, older matches are discarded, and new matches are incorporated into the training set. This procedure simulates realistic forecasting scenarios, where predictions are always made using only past information.

This validation strategy is particularly critical in football analytics for two reasons. First, team performance evolves over time due to tactical adjustments, injuries, and contextual factors. Second, ignoring temporal dependencies can lead to information leakage and overly optimistic performance estimates. Recent research strongly recommends temporal validation for sports analytics models to ensure realistic and reliable evaluation (Herold et al., 2019).

By integrating sliding-window validation directly into the Rulex Factory workflow, the project ensures that model performance reflects real-world applicability rather than static, retrospective fitting

4.4 Explainable Artificial Intelligence

4.4.1 Definition and Motivation

Explainable Artificial Intelligence (XAI) refers to a set of methods and design principles aimed at making the behaviour, decisions, and outcomes of artificial intelligence systems understandable to human users. According to the IEEE, XAI encompasses techniques that enable users to comprehend, trust, and effectively manage AI systems by providing transparent, interpretable, and justifiable explanations of model outputs (Longo et al., 2025). In contrast to traditional black-box models, XAI seeks to reveal *why* a model produces a particular prediction, *which inputs are most influential*, and *how changes in inputs affect outcomes*.

The importance of XAI has grown significantly in applied domains where decisions carry strategic, economic, or ethical implications. In sports analytics and football analytics in particular, model outputs are often used to inform tactical decisions, performance evaluations, and long-term planning. In such contexts, predictive accuracy alone is insufficient. Analysts and coaches must be able to interpret model reasoning in football-specific terms, assess whether conclusions are plausible, and integrate insights with expert knowledge. As a result, explainability is not merely a desirable property but a functional requirement.

From a methodological perspective, XAI is especially relevant when working with limited and noisy datasets, as is the case in football analytics. Complex models may fit training data well but rely on spurious correlations or unstable patterns. Explainable methods help identify these issues by exposing the relationships learned by the model, thereby improving robustness, trust, and scientific validity (Ribeiro et al., 2016; Interpretable Machine Learning - Christoph Molnar - Google Books, n.d.).

4.4.2 Taxonomy of Explainable AI Methods

XAI methods can be broadly categorised according to when and how explanations are generated. A commonly adopted distinction separates intrinsic (model-based) explainability from post-hoc explainability, with further subdivisions based on the scope and granularity of explanations.

Intrinsic (Inherently Interpretable) Models

Intrinsic XAI methods rely on models whose structure is interpretable by design. Examples include linear models, decision trees, rule-based classifiers, generalized additive models (GAMs), and logic-based learners. In these models, the relationship between input features and outputs is directly observable, allowing explanations to be derived without auxiliary tools.

In the context of this thesis, models such as Decision Trees, Explainable Boosting Machines, and the Logic Learning Machine fall into this category. Their interpretability makes them particularly suitable for football analytics, as learned rules or feature effects can be directly translated into tactical interpretations.

Post-hoc Explainability Methods

Post-hoc methods aim to explain the behaviour of already trained black-box models. These approaches do not alter the underlying model but instead provide explanations after predictions have been generated. Post-hoc methods can be further divided into:

- **Global explanations**, which describe overall model behaviour across the dataset;
- **Local explanations**, which focus on individual predictions.

Popular post-hoc techniques include feature attribution methods (e.g., SHAP, permutation importance), surrogate models, and sensitivity analysis (Lundberg & Lee, 2020).

While post-hoc methods are powerful, they introduce an additional layer of approximation, as explanations are not guaranteed to faithfully represent the true internal logic of the model. For this reason, they are often complemented—or replaced—by intrinsically interpretable approaches in domains where trust and transparency are critical.

4.4.3 Feature Importance as a Core Explainability Mechanism

Among the various XAI techniques, feature importance methods play a central role in explaining machine learning models, particularly in tabular, structured-data settings such as football analytics. Feature importance aims to quantify the contribution of each input variable to the model's predictions, thereby identifying which parameters are most influential.

Feature importance methods are especially valuable in football analytics for two reasons. First, they directly address the core research question of this thesis: understanding the importance of performance parameters in shaping team outcomes. Second, they produce explanations that are easily interpretable by domain experts, as features often correspond to intuitive tactical concepts (e.g., pressing intensity, passing behaviour, spatial distribution of actions).

Model-Specific Feature Importance

Some models provide feature importance measures as part of their internal structure. For example, decision tree-based models estimate importance based on the reduction in impurity achieved by splits involving each feature. Ensemble models aggregate these reductions across trees.

While such measures are computationally efficient, they can be biased toward features with many possible split points or higher variance. Nevertheless, when interpreted cautiously, they provide useful global insights into model behaviour.

Model Agnostic Feature Importance

Model-agnostic approaches assess feature importance independently of the model's internal mechanics. One widely used technique is permutation importance, which measures the decrease in model performance when the values of a feature are randomly permuted. A large performance drop indicates that the model relies heavily on that feature.

Another prominent approach is Shapley-based methods, which allocate importance scores based on cooperative game theory principles. These methods provide theoretically grounded explanations but are computationally more demanding (Lundberg et al., 2020).

Feature Importance in Explainable Rule-Based Systems

In rule-based models, feature importance emerges naturally from the structure of the learned rules. Features that appear frequently in high-coverage or high-accuracy rules can be interpreted as more influential. This form of importance is particularly transparent, as it links directly to explicit logical conditions.

In the present project, feature importance is derived through ROC–AUC-based evaluation and through the frequency and positioning of features within Logic Learning Machine rules. This dual approach combines statistical relevance with semantic interpretability, ensuring that importance measures reflect both predictive power and tactical meaning.

4.4.4 Role of Explainable AI in This Study

In this thesis, Explainable AI is not treated as a secondary analytical step but as a guiding principle shaping the entire modelling pipeline. Feature engineering, model selection, and validation strategies are all designed to support interpretability. Feature importance methods, in particular, serve as the primary mechanism for linking machine learning outputs to football knowledge.

By prioritising feature importance–driven explanations, the study enables a structured understanding of team behaviour, highlights which tactical indicators matter most, and provides insights that can be critically evaluated by human experts. This approach aligns with recent recommendations in sports analytics research, which emphasise that explainability is essential for translating data-driven models into actionable and trustworthy insights.

4.5 Summary

This chapter has detailed the complete technical stack and tools used in the thesis. By combining multiple machine learning models within the Rulex Factory and embedding Explainable AI principles throughout the pipeline, the study provides a rigorous, interpretable framework for analysing football team structure under realistic data constraints.

Capture 5

Experimental Results

5.1 Train–Test Splitting Strategy

5.1.1 Motivation for Temporal Splitting in Football Analytics

In supervised machine learning, the design of the train–test split is a critical methodological choice that directly influences the validity of experimental results. In football analytics, this choice is particularly sensitive due to the temporal and contextual nature of match data. Matches are not independent and identically distributed observations: team strength, tactical approaches, injuries, transfers, and psychological factors evolve over time. Consequently, random splitting strategies commonly used in static datasets can introduce information leakage, leading to overly optimistic performance estimates.

Recent research in sports analytics strongly discourages random cross-validation for time-ordered data, advocating instead for temporally consistent evaluation protocols that reflect real-world forecasting scenarios (Herold et al., 2019; Goes et al., 2021). In practical football applications, analysts always predict future matches using past information; therefore, experimental design should mirror this constraint.

For these reasons, the present study adopts a temporal train, test splitting strategy, implemented through sliding-window validation within the Rulx Factory environment.

5.1.2 Sliding-Window Temporal Validation

The experimental framework employs a sliding-window approach, where the training and testing sets are defined over consecutive time intervals rather than randomly sampled observations. At each iteration:

- The training set consists of matches from a fixed historical window.
- The test set consists of matches from the immediately following time window.
- After each iteration, the window slides forward in time: older matches are removed from training, and newer matches are added.

Formally, let $W_t^{\{train\}} = \{m_t, \dots, m_{t+k}\}$ denote the training window and $W_t^{\{test\}} = \{m_{t+k+1}\}$ the test window. At the next iteration, the windows become:

$$W_{t+1}^{train} = \{m_{t+1}, \dots, m_{t+k+1}\}, \quad W_{t+1}^{test} = \{m_{t+k+2}\}.$$

This procedure ensures that:

1. Only past information is used to predict future outcomes.
2. Model performance is evaluated under realistic deployment conditions.
3. Temporal dependencies and non-stationarity are explicitly accounted for.

5.1.3 Single-Competition vs Multi-Competition Splits

Two complementary experimental settings are used:

1. Single Competition Sliding Window

The model is trained and evaluated within a single league (e.g., Serie A). This setting controls for stylistic and structural differences across competitions and provides insight into league-specific patterns.

2. 380-Competitions Sliding Window

The model is trained on multiple competitions with identical season lengths (Premier League, Serie A, La Liga, Ligue 1). A competition identifier is included as a feature, allowing the model to learn both shared and league-specific patterns.

This dual setup allows the analysis of generalisation across leagues while preserving temporal consistency, a methodological choice aligned with best practices in football analytics research (Rein & Memmert, 2016)

5.2 Evaluation Metrics Description

5.2.1 Rationale for Metric Selection

The evaluation of machine learning models in football analytics must balance predictive accuracy, class imbalance, and interpretability. Match outcomes are inherently stochastic, and accuracy alone may not capture a model's ability to discriminate between outcomes across varying thresholds.

For this reason, the study adopts a multi-metric evaluation strategy, focusing primarily on:

- Accuracy
- Area Under the ROC Curve (AUC)

These metrics are widely used in sports analytics and classification tasks and provide complementary perspectives on model performance (Bransen & Van Haaren, 2019)

5.2.2 Accuracy

Accuracy measures the proportion of correctly classified instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Accuracy is intuitive and easy to interpret; however, it can be misleading in the presence of class imbalance or when prediction confidence varies significantly. For this reason, accuracy is reported alongside AUC rather than used as the sole performance indicator.

5.2.3 Area Under the ROC Curve (AUC)

The AUC measures a model's ability to discriminate between classes across all possible decision thresholds. It represents the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance.

An AUC of:

- 0.5 indicates random performance,
- 1.0 indicates perfect discrimination.

AUC is threshold-independent and particularly suitable for football analytics, where decision thresholds may vary depending on tactical or strategic considerations (Herold et al., 2019)

5.2.4 Train vs Test Metrics

For each model and experimental configuration, both training and test metrics are reported. The comparison between these values provides insight into:

- model generalisation,
- susceptibility to overfitting,
- stability across temporal splits.

Large discrepancies between training and test performance are interpreted as indicators of model overfitting or insufficient regularisation.

5.3 Experimental Results

5.3.1 Overview of Models Compared

The experimental evaluation includes the following models, as reported in the Comparative Evaluation Across All Models:

- Decision Tree
- Random Forest
- XGBoost
- Explainable Boosting Machine (EBM)
- Logic Learning Machine (LLM), evaluated under:
 - Single Competition
 - 380-Competitions

Table 5.1 reports the comparative evaluation across all machine learning models considered in this study, including classical tree-based methods, ensemble approaches, and interpretable models. All models were trained on the same feature space and evaluated using identical temporal sliding-window validation schemes, ensuring a fair and consistent comparison.

Table 5.1 Comparative evaluation across all machine learning models using a common 70/30 random train and test split, reporting training and test accuracy, AUC, and top-ranked features.

	Accuracy		AUC		Feature Importance (Top5)	
	Train	Test	Train	Test		
Decision Tree	0.619	0.637	0.765	0.763	crosses_def_home	0.1849
					smart_passes_away_pct	0.1016
					high_passes_def_home	0.0968
					crosses_att_away	0.0921
					simple_passes_att_away_pct	0.0543
Random Forest	0.869	0.864	0.961	0.962	crosses_def_home	0.0374
					crosses_att_away	0.0354
					high_passes_att_away	0.0323
					smart_passes_away_pct	0.0303
					high_passes_def_home	0.0287
XGBoost	1.0000	0.593	1.0000	0.734	crosses_def_home	0.032991
					crosses_att_away	0.032276
					smart_passes_home_pct	0.029507
					high_passes_def_home	0.026102
					simple_passes_att_away_pct	0.025318
Explainable Boosting Machine	0.6671	0.6123	0.82	0.75	crosses_def_home	0.1332
					crosses_att_away	0.1324
					high_passes_def_home	0.1042
					high_passes_att_away	0.0710
					crosses_def_away	0.0675
Logic Learning Machine Single Competition	0.842	0.719	0.91	0.75	smart_passes_home_pct	0.500
					crosses_def_home	0.299
					high_passes_def_home	0.225
					smart_passes_away_pct	0.106
					high_passes_att_away	0.086

Why we use LLM:

The Logic Learning Machine (LLM) is included in the experimental evaluation to address the central methodological objective of this study: combining competitive predictive performance with full model interpretability. Unlike black-box ensemble methods, LLM generates explicit IF–THEN rules that directly link tactical performance indicators to match outcomes, enabling transparent inspection of model reasoning. This property is particularly important in football analytics, where understanding why certain team behaviours influence results is as critical as predictive accuracy. As shown in Table 5.1, LLM achieves competitive AUC values with a more balanced train–test behaviour than highly expressive models such as XGBoost, indicating stronger robustness under limited and temporally structured data. Moreover, the rule-based structure of LLM allows feature importance to emerge naturally through rule frequency and coverage.

5.3.2 Performance Summary Across Models

The comparative tables indicate a clear stratification of models in terms of performance and generalisation:

- **Decision Tree** models show modest accuracy and AUC, reflecting their limited capacity and high variance.
- **Random Forest** achieves strong and stable performance on both training and test sets, with minimal degradation, indicating robust generalisation.
- **XGBoost** exhibits near-perfect training performance but significantly lower test performance, revealing pronounced overfitting.
- **EBM** achieves intermediate performance, outperforming simple trees while maintaining interpretability.
- **LLM** demonstrates competitive AUC values with relatively balanced train–test behaviour, especially in the single-competition setting.

Across all models, zone-based passing and crossing features consistently emerge among the most influential predictors

5.3.3 Logic Learning Machine Results

After the global comparison across all models, this section focuses on the detailed experimental results obtained with the Logic Learning Machine (LLM), which represents the core explainable model of this study.

In the Single Competition, LLM achieves higher accuracy and AUC compared to the multi-competition setting, reflecting the relative homogeneity of tactical patterns within a single league.

In the 380-Competitions, performance decreases moderately, indicating the increased difficulty of generalising across leagues with distinct playing styles. However, the gap between training and test performance remains smaller than that observed for XGBoost, suggesting better robustness.

Importantly, the extracted rules consistently involve:

- zone-based passing distributions,
- defensive action frequencies,
- crossing behaviour.

These results confirm that LLM is not only predictive but also capable of capturing structural tactical regularities.

Table 5.2 Logic Learning Machine performance under the Sliding Window evaluation, reporting average training and test AUC and accuracy for different feature set sizes.

Evaluation Type(5 Attributes)	Competitions Considered	Average AUC	Accuracy
Single Competition Sliding Window	Serie A (Italy)	Train: 0.825 Test: 0.804	0.63
380 Competitions Sliding Window	Premier League, Serie A, La Liga	Train: 0.774 Test: 0.713	0.53

Evaluation Type(10 Attributes)	Competitions Considered	Average AUC	Accuracy
Single Competition Sliding Window	Serie A (Italy)	Train: 0.843 Test: 0.783	0.65
380 Competitions Sliding Window	Premier League, Serie A, La Liga	Train: 0.782 Test: 0.734	0.56

Evaluation Type(20 Attributes)	Competitions Considered	Average AUC	Accuracy
Single Competition Sliding Window	Serie A (Italy)	Train: 0.843 Test: 0.783	0.66
380 Competitions Sliding Window	Premier League, Serie A, La Liga	Train: 0.793 Test: 0.734	0.57

Evaluation Type(70 Attributes)	Competitions Considered	Average AUC	Accuracy
Single Competition Sliding Window	Serie A (Italy)	Train: 0.87 Test: 0.762	0.68
380 Competitions Sliding Window	Premier League, Serie A, La Liga	Train:0.80 Test: 0.723	0.59

5.4 Discussion

5.4.1 Impact of Train–Test Splitting Strategy

The use of sliding-window validation proves essential for realistic performance assessment. Models evaluated using temporally consistent splits exhibit lower but more reliable performance than would be expected under random cross-validation. This finding aligns with prior work demonstrating that temporal leakage can substantially inflate performance estimates in sports analytics (Goes et al., 2021)

5.4.2 Predictive Performance vs Interpretability Trade-Off

The experimental results clearly illustrate the trade-off between predictive performance and interpretability:

- Ensemble methods (Random Forest, XGBoost) achieve high accuracy but offer limited transparency.
- Interpretable models (Decision Tree, EBM, LLM) provide actionable insights but slightly lower raw performance.

Among interpretable models, LLM stands out by achieving competitive AUC while preserving full semantic transparency through rule-based explanations.

5.4.3 Overfitting and Model Complexity

The severe overfitting observed in XGBoost highlights the risks of deploying highly expressive models in football analytics under limited data regimes. Despite regularisation, the model captures noise and league-specific artefacts rather than generalisable patterns.

In contrast, LLM and EBM exhibit more controlled complexity, leading to better generalisation and more stable temporal behaviour.

5.4.4 Tactical Insights and Feature Importance

The consistency of top-ranked features across models, particularly zone-based passing and crossing metrics, supports the validity of the feature engineering pipeline described in Chapter 3. These features encode meaningful tactical information and appear robust across leagues and modelling approaches.

The alignment between statistical importance (AUC ranking) and logical importance (LLM rule structure) strengthens confidence in the interpretability of the results.

5.4.5 Limitations and Future Directions

While the experimental results are encouraging, several limitations remain. The dataset size constrains the exploration of more complex temporal models, and the focus on event data excludes off-ball dynamics. Future work may integrate tracking data or extend the temporal horizon across multiple seasons to further enhance generalisation.

Conclusion

This thesis set out to investigate football team structure and the relative importance of performance parameters through an explainable machine learning framework grounded in event-based data. Motivated by the growing adoption of data-driven decision-making in professional football and the parallel demand for transparency and interpretability, the study was designed to move beyond purely predictive modelling and toward knowledge extraction that is both methodologically rigorous and tactically meaningful. By integrating data engineering, domain-informed feature construction, temporal evaluation, and explainable modelling within a unified workflow, the work contributes to the ongoing discourse on how machine learning can be responsibly and effectively applied in sports analytics.

From a data perspective, the thesis demonstrated the importance of a structured and reproducible acquisition and integration pipeline. Raw football event data, while rich in detail, is inherently hierarchical and not directly suitable for machine learning analysis. The transformation of nested JSON files into a relational SQLite database enabled consistent aggregation, transparent feature extraction, and traceability across matches, teams, and competitions. This database-centric design proved essential for supporting downstream analytical tasks and aligns with best practices in contemporary football analytics research, where reproducibility and data governance are increasingly emphasised.

A central contribution of the work lies in the feature engineering strategy. Rather than relying on automatically learned representations or deep learning approaches that are ill-suited to limited and noisy datasets, the thesis adopted a feature-centric methodology rooted in football theory. Aggregated statistical features, tactical indicators such as PPDA and xG, and zone-based spatial abstractions collectively formed a compact yet expressive representation of team behaviour. The empirical results confirmed that these features consistently carried explanatory power across models and competitions, supporting the argument that domain-informed features can outperform more complex representations when interpretability and robustness are primary objectives.

The modelling and evaluation phase further reinforced the necessity of aligning methodological choices with the realities of football data. Through the use of sliding-window temporal validation, the study avoided information leakage and provided performance estimates that more closely reflect real-world deployment scenarios. The comparative evaluation across multiple machine learning models highlighted clear trade-offs between predictive performance and interpretability. While ensemble methods such as Random Forests and Gradient Boosting achieved strong accuracy, they exhibited limitations in transparency and, in some cases, susceptibility to overfitting. In contrast, interpretable models, particularly the Logic Learning Machine, offered a balanced compromise, achieving competitive discriminatory power while producing explicit, human-readable rules that directly map to tactical concepts.

The Logic Learning Machine emerged as a particularly valuable tool for the research objectives of this thesis. Its rule-based structure enabled the extraction of semantically meaningful patterns linking combinations of passing behaviour, defensive actions, and spatial distributions to match outcomes. Importantly, the consistency between statistically ranked features and those appearing most frequently in high-coverage rules strengthens confidence in the validity of the findings. This dual perspective quantitative feature importance, complemented by logical explanations, demonstrates how explainable AI can bridge the gap between data-driven modelling and football expertise, a challenge frequently identified in the literature.

From a football analytics standpoint, the results provide evidence that certain structural aspects of team play are robustly associated with success across contexts. Zone-based passing distributions, crossing behaviour, and indicators of pressing intensity repeatedly emerged as influential features, suggesting that spatial control and off-ball defensive organisation are central components of effective team performance. At the same time, the observed performance differences between single-competition and multi-competition settings underline the contextual nature of football tactics. While some patterns generalise across leagues, others are shaped by competition-specific styles and constraints, highlighting the importance of contextualised analysis rather than one size fits all models.

Despite these contributions, the thesis also acknowledges several limitations. The reliance on event-based data excludes off-ball movements and spatial interactions that are only fully captured by tracking data. Additionally, the analysis is constrained to the available temporal scope and competition set, which may limit the exploration of long-term tactical evolution. These limitations, however, do not detract from the core methodological message of the

study. Instead, they point toward clear directions for future research, including the integration of tracking data, multi-season analyses, and hybrid frameworks that combine interpretable models with richer spatial representations while preserving transparency.

In conclusion, this thesis demonstrates that explainable machine learning provides a viable and scientifically grounded approach to analysing football team structure and performance determinants. By prioritising interpretability, temporal validity, and domain knowledge, the proposed framework delivers insights that are not only statistically sound but also actionable and communicable to practitioners. As football continues to embrace data-driven methods, such approaches will be increasingly important to ensure that analytical models support, rather than obscure, tactical understanding. The findings of this work, therefore, contribute both to the academic literature on sports analytics and to the practical development of transparent decision-support tools in professional football.

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